



Can we Conceive of an Alternative to the Electric Car?

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I declare that the work in this dissertation was carried out in accordance with the requirements of the University's Regulations and Code of Practice for Taught Programmes and that it has not been submitted for any other academic award. Except where indicated by specific reference in the text, this work is my own work. Work done in collaboration with, or with the assistance of others, is indicated as such. I have identified all material in this dissertation which is not my own work through appropriate referencing and acknowledgement. Where I have quoted or otherwise incorporated material which is the work of others, I have included the source in the references. Any views expressed in the dissertation, other than referenced material, are those of the author.

Abstract

NIFTI (National Individual Floating Transport Infrastructure) is a repulsive-roadway maglev concept in which an autonomous, permanent-magnet pod is levitated above a bed of individually driven electromagnet coils. By removing contact with the road, the concept could avoid tyre and road wear, which are important sources of particulate emissions from road transport. Because the pod carries a fixed magnetic moment, Earnshaw's theorem prevents a stable static equilibrium, meaning the pod sits atop a saddle point, with restoring behaviour vertically but anti-restoring behaviour laterally. Continuous active control is therefore required to keep the pod at this equilibrium point.

This work began with a single 1,861-line simulation script with no test suite and an analytically fitted, position-only current assignment method for the coils instead of a more advanced control system. The script was reconstructed into a modular codebase of seven dependency-ordered packages containing 33 modules across seven packages. The reconstruction was carried out incrementally while preserving behaviour, with per-function provenance tracking, per-package regression tests, and a runtime comparison against the original implementation using sweeps rather than CLI. The reconstructed model reproduces all twelve pod state components of the original through the onset of instability. Three extensions were then added: a closed-loop control layer separating force generation from current allocation, a unit-quaternion/SO(3) attitude integrator, and a multi-dipole replacement to the single-dipole force model.

The primary finding of the whole work is that the inherited current schedule cannot levitate the pod and cannot be made to by any control law. The reconstruction made this experimentally verifiable. Its raw lift is hundreds of times below the pod weight, while its lateral effects are destabilising, leaving no stable operating point for the controller. The point-dipole pod model can produce five of the six required force directions, but cannot generate torque about the magnetic moment axis. The multi-dipole model removes this limitation and provides full six-degree-of-freedom authority through the same coils, although pitch is substantially more expensive to produce. The SO(3) integrator removes the problem of Euler-angle discontinuity through large angular rotations while agreeing with the inherited model to within a milliradian near level attitude(10°). With the height actively held, the lateral PD control drives a 6.4 mm offset below 1 mm within about 70 ms, displaying lateral authority, however is unable to maintain positional stability over the full 300 ms using the un-tuned control law.

The result is therefore not a completed levitating, stationary controller. But the potential of the constructed system in which the limitations of the inherited model can be separated from the behaviour of the new control systems and modelling components. Establishing an accurate levitation baseline, supported by physical rig data, is the next step needed before the controller and six-degree-of-freedom model can be tested as a complete system.

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Chapter 1

Background

This chapter provides the background needed for the rest of the dissertation. §1.1 introduces the transport problem that NIFTI is intended to address. §1.2 describes the NIFTI pod-and-track concept and its project lineage. §1.3 introduces the magnetic theory needed to understand how the pod is supported and disturbed, covering the force and torque acting on a magnetic dipole and using this to distinguish the two established families of maglev suspension, of which NIFTI is one.

§2.4 then applies this theory to the specific pod-track model used by the simulation. It describes the coil-field kernel, the dipole approximations used for the pod and their error bounds, and the rigid-body dynamics used to simulate its motion. §1.5 develops and proves Earnshaw's theorem, showing why the pod's equilibrium is a saddle point. It also shows why the pod's orientation matters as well as its position, and why one rotational axis behaves differently from the other two. §1.6 The current state of the control mechanism in the simulation, with §1.7 reviewing approaches used to actively stabilise this type of unstable magnetic equilibrium in the control-engineering and maglev literature. Finally, §1.8 identifies the gaps addressed by this work, in their architectural . These gaps motivate the objectives in Chapter 1 and the work presented in Chapter 2 and onwards.

1.1 Motivation

Road transportation is the most-used form of transport in the UK¹, but it carries a cost far beyond its direct convenience. Three such issues are worth separating, as each motivates a different requirement for a replacement technology.

Congestion. Time and productivity lost to traffic congestion is large and quantifiable, and is not evenly distributed across road networks: it is concentrated specifically in dense, start-stop urban environments. Across Great Britain, delay on local A roads averages 74.1 seconds per vehicle per mile in urban areas, against 21.5 seconds per mile in rural areas, this is more than three times the delay, at little over half the average speed (17.6 mph urban versus 34.4 mph rural)². That is a clear trend country wide however, it is noticeably worse in London, with its drivers losing 101 hours to congestion in 2024, causing a speculated national cost of £7.7 billion³. Congestion is not solved by changing what powers a vehicle; it is a consequence of road demand exceeding road-space supply in towns, cities and motorways where that demand is highest and existing road geometry is most constrained, with any technology addressing it having to either use existing road space more efficiently or overcome the shared, contact-based road surface altogether. Neither is this simply a matter of insufficient capacity: added road

capacity itself induces the demand that fills it, so widening urban roads does not durably relieve congestion⁴

Land use. Road infrastructure: carriageways, junctions, and parking consumes a substantial and largely irrecoverable share of urban land, competing directly with housing, green space, and other productive uses in precisely the areas where transport demand is highest. This is amplified in cities where car-centric infrastructure dominates public space, further driving the need for private cars in daily life⁵.

Safety. It remains unclear whether autonomous control alone does - or rather will - improve safety outcomes relative to human drivers. But what is clear is the effect of mass on severity of incidents: kinetic energy scales with mass, so lighter weight cars lacking an engine or automotive-scale battery packs are substantially lighter than the vehicles they replace, and a lighter vehicle population produces less severe collisions when they do occur⁶. This benefit is conditional rather than universal: NHTSA's own analysis of the US fleet found mass reduction is protective mainly where it narrows the disparity between colliding vehicles, and can increase risk to a lighter vehicle's own occupants in a mixed fleet that includes much heavier vehicles⁷.

Emissions and particulates. Traditional motor vehicles produce large amounts of particulate emission, with the health impacts of this well recorded for a long period of time. Electrification was signalled to be the solution to this however, whilst it addresses tailpipe emissions, it does not eliminate particulate pollution produced via surface wear of tyres and roads. This class of particulate is dubbed PM₂₋₅ and exist within a micrisocipic scale that has profound health and social impacts⁸. In fact the additional weight of EV battery packs increases non-exhaust particulate emissions from tyre and brake wear to a level that can exceed tailpipe emissions from an equivalent combustion vehicle⁹. This means that a fully electric car can still contribute more particulate pollution, via wear rather than exhaust, than the vehicle it replaced. Using 2021 UK National Atmospheric Emissions Inventory data, road-transport PM_{2.5} splits roughly 24% exhaust to 76% non-exhaust (52% tyre and brake wear, 24% road abrasion)¹⁰.

It is clear from looking at these features that the current trajectory of solution that is being targetted - that being a shift towards EV's, does not inherently eliminate many of the root problems present in urban areas when it comes to traditional means of transport.¹¹⁻¹³ Taken together, these costs point toward one ideal solution: a technology that removes physical wheel-road contact entirely (addressing particulates at the source, not just tailpipe emissions), that changes the geometry of how vehicles occupy space (addressing congestion and land use), and that reduces total vehicle mass (furthering the safety and energy benefits mentioned), rather than one that simply replaces the propulsion source of an otherwise-unchanged vehicle¹⁴.

1.1.1 Alternative Transport Concepts

Many existing transport concepts address parts of the requirements above, but neither fully addresses the combined problem considered by NifTI. **Personal Rapid Transit (PRT)**, including the ULTra system at Heathrow Airport and the Morgantown network, operating since 1975¹⁵, provides individually routed, on-demand transport on dedicated guideways, closely matching NifTI's individual pod transit model. However, these systems remain dependent on contact-based mechanical technologies, retaining the associated wear and particulate emissions of this category¹⁶. **Hyperloop**, by contrast, combines maglev-derived levitation with reduced aerodynamic drag for ultra-high-speed intercity transport¹⁷, sharing physical similarities with NifTI at the suspension level but not at the system level; its enclosed tube infrastructure is poorly suited to individually routed, stop-start, junction-heavy urban movement, and its practical maturity remains unclear and contested. Together, these concepts illustrate the gap NifTI seeks to address: combining PRT's individually routed, junction-capable operation with contact-free

vehicle suspension at urban scale.

1.2 The NifTI Project

Magnetic levitation (maglev) is the family of technology that meets the contact-removal requirement previously described directly: a vehicle supported and guided entirely by magnetic force, with no mechanical contact with its guideway at standstill or at speed. Successful maglev systems have carried passengers for decades on a range of suspension mechanisms and are increasingly more sought after solutions.

1.2.1 The Pod-and-Track

NifTI (National Individual Floating Transport Infrastructure) is a roadway-based maglev concept: wherein a centrally-coordinated fleet of autonomous pods, each constituted of a permanent magnetic base, running above repulsive magnetic coils buried in the roadway¹⁸. These coils when driven with a controlled current would repel the pod's permanent magnets, holding it above the track.

The coils (later referred to as actuators of the lift force) would be arranged in transverse rows of vertical-axis cylindrical electromagnet coils, spaced along a 2D plane along the direction of travel and distributed across the track width; the pod carries its permanent magnets in vertical stacks in its base. The specific details of these configurations is provided in methods.

1.2.2 Origins and Development History

NifTI was invented and patented in 2019 by Professor Nigel Hussey (Universities of Bristol and Radboud), in collaboration with the University of Applied Sciences Arnhem and Nijmegen (HAN)¹⁸. The project is at its practical experimental inception, with the only experiment taking place having been a toy example pod on a 1 m long track, which inspired the development of the simulation code. The code has been a multi-generation lineage of University of Bristol students that have progressively developed the code. This code implements the pure physics of the system with computer science students at Radboud working on the more traditional central control routing problems. This dissertation continues the physics programme directly.

1.2.3 The Inherited Software and Data

The immediately preceding contribution, Holman's dissertation¹⁹, delivered the materials this work builds on directly: a working simulation of a levitated pod above a coil track; and an empirical, position-dependent current schedule intended to hold the pod near its equilibrium. Both came as a single, unmodularised script, preserved unmodified in the repository as `base_line_code.py`.

Track geometry is constructed via inherited files(.xlsx) detailing preset coil locations and constituent currents which were recorded experimentally on the toy rig. Additionally there was a suite of different pod geometries, and one bench recording of the magnetic field at eight sensor positions above the track.

1.3 Magnetic Dipole Forces, and the Types of Suspension

The physics of the simulated system are explained here.

1.3.1 Force and Torque on a Magnetic Dipole

A magnetic field $\mathbf{B}$ is produced by electric current; for the cylindrical coil windings used here it is evaluated from a closed-form finite-source solution, as derived in, Appendix *Field-Kernel Implementation Details*). A body whose net magnetic response is captured by a single dipole moment $\mathbf{m}$ interacts with an external field through two expressions²⁰:

$$\boldsymbol{\tau} = \mathbf{m} \times \mathbf{B}, \quad (1.1)$$

$$\mathbf{F} = \nabla(\mathbf{m} \cdot \mathbf{B}), \quad (1.2)$$

a torque that acts to align $\mathbf{m}$ with $\mathbf{B}$, and, in a no current region, a net force which is equal to $(\mathbf{m} \cdot \nabla)\mathbf{B}$. This results in:

Force comes from field gradients. $\nabla(\mathbf{m} \cdot \mathbf{B}) = 0$ when $\mathbf{B}$ is spatially uniform, a uniform field exerts a torque but no net force. This means a maglev system's lift and guiding forces are proportional to the field gradient, not its strength.

Torque is perpendicular to the moment. $\boldsymbol{\tau} = \mathbf{m} \times \mathbf{B}$ has no component along $\mathbf{m}$, for *any* field and it vanishes when $\mathbf{m} \parallel \mathbf{B}$. Torque about the dipole's own axis is therefore structurally unreachable through the field.

A body well approximated by one dipole moment is fully described, for both force and torque, by Equations (1.1)–(1.2).

The suitability of these assumptions is examined further in the methods and results applied to this simulation code.

1.3.2 Electromagnetic Induction

An alternative to driving current directly is to have it induced via motion. A conductor moving through a spatially varying field sees a changing flux, which owing to Faraday's law drives an electromotive force and circulating eddy currents that oppose the change (this being Lenz's law). This means that their own induced field repels the magnet. Because the induced current is set by the *rate* of flux change, it scales with speed: at standstill there is no current and no lift. This is the entire lift mechanism of electrodynamic suspension, and the reason it has a threshold speed[?]. Electromagnetic suspension does not use these features: instead its lift is actively driven.

1.3.3 Electromagnetic and Electrodynamic Suspension

The two established maglev suspension families are distinguished by which of the above they rely on²¹.

Electromagnetic suspension (EMS) supplies current directly to the coils; the resulting lift force (Equation (1.2)) is available across all speeds, including standstill, as it depends only on the driven field gradient, not on relative motion or a changing flux. Transrapid, the most successful

implementation, lifts by *attraction* towards an overhead rail²²; NifTI plans on inverting this mechanism with the *repulsive* roadway configuration¹⁸. The field gradient and thus the force available falls off steeply with distance, meaning an EMS gap is small (of the order 10 mm). However, maintaining this gap is not a passive action. Earnshaw’s theorem(as described in §1.5.1) forbids this. *Instead it must rely on active control.*

Electrodynamic suspension (EDS) uses induction, where superconducting magnets moving over a conductive guideway induce the repulsive eddy-current field, which becomes self-stabilising once the vehicle is fast enough. This results in a much larger gap (~ 100 mm) with no active correction required²³. The majority of mature international programmes: JR-Maglev/SCMaglev, the Inductrack concept, are all based in EDS^{24–26}. However, its principle drawback follows from the necessity for the eddy current dynamics, where below a threshold speed the repulsion is too weak to support the vehicle, so EDS vehicles need wheels for standstill and low-speed.

	EMS	EDS
Mechanism	Direct magnetic attraction/ repulsion using electromagnets	Induced currents in conductive coils due to relative motion
Force at standstill	Available; independent of vehicle speed	None; levitation requires relative motion and sufficient speed
Typical levitation gap	~ 10 mm	~ 100 mm
Stability	Intrinsically unstable; requires continuous active control	Inherently stable once sufficient speed is reached
Representative systems	Transrapid ²²	JR-Maglev, SCMaglev, Inductrack ^{24–26}

Table 1.1: Comparison of electromagnetic suspension (EMS) and electrodynamic suspension (EDS) systems.

From this it is clear that EDS is ruled out by its zero-speed limitation. What the commitment requires is coil currents varied in real time to hold an equilibrium. Continuous active control is therefore the central engineering problem the later chapters address.

1.4 The Simulation Model

§2.3 gave the general dipole force and torque equations. This section covers the specific model the simulation uses, the coil field, how the pod is represented, and the rigid-body equations it follows. All outputs are as a result of the set of modelling assumptions rather than real physically collected rig data.

1.4.1 The Coil-Field Kernel

Every magnetic source from the permanent-magnet pod stack to the current-carrying track coils are modelled as a solid cylinder with uniform magnetisation, and its field is worked out, in closed form, using complete elliptic integrals, following Derby & Olbert²⁷. This is a standard result in electromagnetism²⁸. It avoids the error and cost of numerically integrating the Biot–Savart law over each coil’s surface. The details of how the scheme works specifically are discussed in §?? and the Bulirsch cel routine are in Chapter 3, Appendix *Field-Kernel Implementation Details*.

This can be simplified as: the whole collection (track or pod) is solely the vector sum of the single-cylinder result over its magnets, growing linearly with the number of terms. Next the only quantity in the solution that depends on current, $B_0 = \mu_0 M / \pi$ (with $M = NI\mu_r / L$ for a

coil of N turns, current I , and length L), multiplies the whole field. So a coil's field/ force is proportional to that coil's current.

1.4.2 The Pod: Dipole Models, Validity, and Potential Energy

The pod's force and torque can be computed in one of two ways

- a **single equivalent dipole** $\mathbf{m}$ at the centre of mass, making it the first term of a Taylor expansion of the field across the pod, valid when the field *gradient* does not change much over the pod's size.
- a **per-magnet multi-dipole** model: where each magnet at its own position, still treated as a point dipole, with the total torque including the moment from the rigid body.

The track field is never a point-dipole approximation, but instead is always treated with the formula. Instead every use of a dipole approximation is in reference to the ferromagnetic pod.

The single-dipole model is the default as it existed before this work. The multi dipole model has unique properties making it worth simulating and extending the code to obtain.

The modelling assumption of treating one magnet as a point dipole is fine when the magnet is near the centre of the track. Merging several magnets into one dipole for the whole pod is a large assumption given the pod's size next to the ~ 10 mm gap. The difference between the single, multi and cloud distributed dipole models is later addressed.

For two dipoles $\mathbf{r} = r\hat{\mathbf{r}}$ apart, Equation (1.2) is:

$$\mathbf{F} = \frac{3\mu_0}{4\pi r^4} \left[(\mathbf{m}_1 \cdot \hat{\mathbf{r}})\mathbf{m}_2 + (\mathbf{m}_2 \cdot \hat{\mathbf{r}})\mathbf{m}_1 + (\mathbf{m}_1 \cdot \mathbf{m}_2 - 5(\mathbf{m}_1 \cdot \hat{\mathbf{r}})(\mathbf{m}_2 \cdot \hat{\mathbf{r}}))\hat{\mathbf{r}} \right]. \quad (1.3)$$

However, a pod of fixed moment $\mathbf{m}$ in *any* external field $\mathbf{B}$ has magnetic potential energy:

$$U = -\mathbf{m} \cdot \mathbf{B}, \quad (1.4)$$

with $\mathbf{F} = -\nabla U = \nabla(\mathbf{m} \cdot \mathbf{B})$ giving back Equation (1.2). Setting $\mathbf{B}$ to the field of a single source dipole $\mathbf{m}_1$ turns Equation (1.4) into

$$U = \frac{\mu_0}{4\pi r^3} \left[\mathbf{m}_1 \cdot \mathbf{m}_2 - 3(\mathbf{m}_1 \cdot \hat{\mathbf{r}})(\mathbf{m}_2 \cdot \hat{\mathbf{r}}) \right]. \quad (1.5)$$

It depends on both distance ($1/r^3$; $1/r^4$ in the force) and relative orientation, that orientation dependence is exactly what the inherited, position-only control method does not track.

1.4.3 Rigid-Body Dynamics

The pod is modelled as a rigid body with twelve states: three for cartesian position $\mathbf{x}$, three for velocities, and the same for $\dot{\mathbf{x}}$, attitude (ϕ, θ, ψ) , and accompanying body angle velocity $\boldsymbol{\omega}$, under:

$$m\ddot{\mathbf{x}} = (\mathbf{m}_{\text{world}} \cdot \nabla)\mathbf{B} - mg\hat{\mathbf{z}}, \quad \boldsymbol{\tau} = \mathbf{m}_{\text{world}} \times \mathbf{B}, \quad I\dot{\boldsymbol{\omega}} = \boldsymbol{\tau}, \quad (1.6)$$

with $\mathbf{m}_{\text{world}} = R(\phi, \theta, \psi) \mathbf{m}_{\text{body}}$ the pod's dipole moment rotated into the world frame by its current attitude, and I its inertia tensor, computed from the magnet geometry. Two inherited approximations also sit inside Equation (1.6). The full rotational equation has an extra gyroscopic term, $I\dot{\boldsymbol{\omega}} = \boldsymbol{\tau} - \boldsymbol{\omega} \times (I\boldsymbol{\omega})$, which is nonlinear in $\boldsymbol{\omega}$ even at constant torque, making it

much harder to model. The body rates are integrated directly as if they were rates of change of the Euler angles.

The body rates are not the rates of change of the Euler angles. For the intrinsic Z-Y-X convention (roll about x , pitch about y , yaw about z ; $R(\phi, \theta, \psi) = R_z(\psi) R_y(\theta) R_x(\phi)$) the two are linked by

$$\begin{pmatrix} \dot{\phi} \\ \dot{\theta} \\ \dot{\psi} \end{pmatrix} = \begin{pmatrix} 1 & \sin \phi \tan \theta & \cos \phi \tan \theta \\ 0 & \cos \phi & -\sin \phi \\ 0 & \sin \phi / \cos \theta & \cos \phi / \cos \theta \end{pmatrix} \boldsymbol{\omega}, \quad (1.7)$$

and the inherited integrator uses the small-angle form of Equation (1.7), in which the matrix becomes the identity and $[\dot{\phi}, \dot{\theta}, \dot{\psi}] = \boldsymbol{\omega}$, this is accurate only while the attitude stays within tenths of a radian (typical small angle approximation). Equation (1.7) collapses at $\theta = \pm\pi/2$, where $\cos \theta \rightarrow 0$: the roll and yaw axes line up, a degree of freedom is lost, and the angles jump instead of changing smoothly. This is known as gimbal lock²⁹. An alternate method for solving this is known as a unit-quaternion / SO(3) integrator which has no such issue and is investigated later.

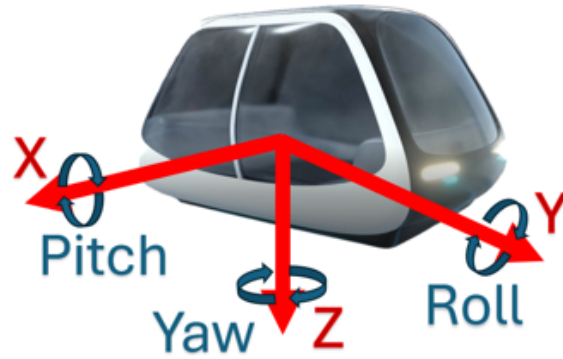


Figure 1.1: The three body-axis rotations: roll about x , pitch about y , and yaw about z , in the intrinsic Z-Y-X convention (roll applied to the body vector first, yaw last) used throughout this dissertation.

1.5 EMS Instability and Orientation

1.5.1 Earnshaw's Theorem and Magnetic Saddle Point

The need for active control follows from the physics described by Earnshaw's theorem³⁰. In a current-free region, Maxwell's equations require the magnetic field to satisfy

$$\nabla \cdot \mathbf{B} = 0, \quad \nabla \times \mathbf{B} = 0,$$

meaning that a static magnetic field cannot form a stable equilibrium for a pod with a fixed magnetic moment in all directions. Instead, any equilibrium behaves as a saddle point: if the field is restoring in one direction, it must be actively destabilising in another²⁰.

This result assumes both a fixed magnetic moment and a static magnetic field. NIFTI avoids the latter through active control, continuously varying the coil currents and therefore the magnetic field in response to the pod's position. For the configurations considered in this dissertation, the nominal equilibrium is restoring vertically but unstable laterally, creating the saddle structure that makes lateral feedback control necessary.

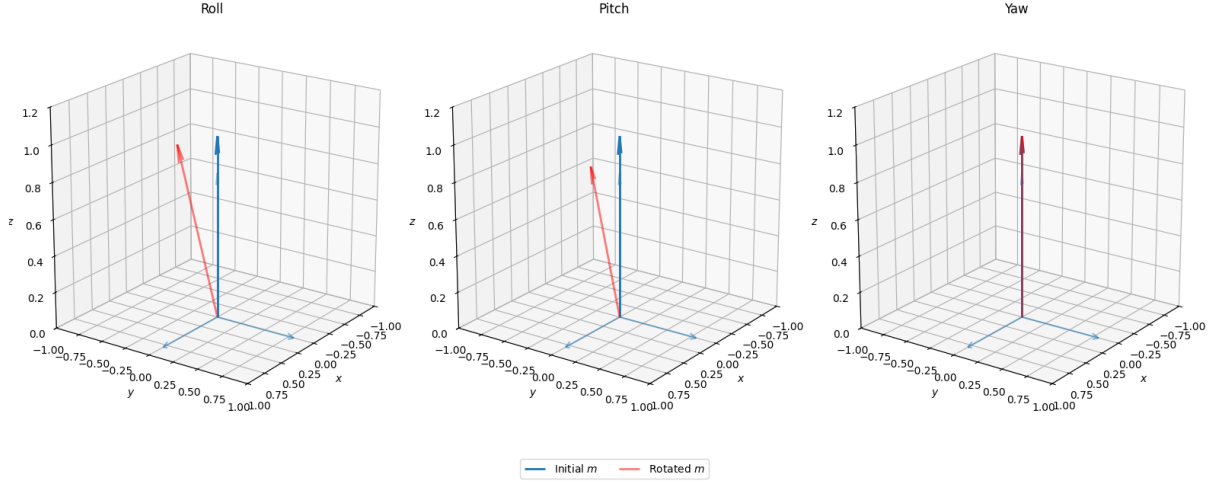


Figure 1.2: The pod’s dipole moment $\mathbf{m}$ (bold arrow) under a small rotation about each Euler axis, against a dashed world reference frame. Roll and pitch tip $\mathbf{m}$ visibly off vertical; yaw rotates the pod about the vector’s own base without changing $\mathbf{m}$ ’s direction at all. Formally, $R_{\mathbf{m}}(\theta) \mathbf{m} = \mathbf{m}$ for rotation about a vector’s own axis, whatever θ ; rotation about any other axis moves it.

1.5.2 Why Orientation Governs the Equilibrium

The saddle’s location and characteristics also depend on the pod’s *orientation*, which is not uniform across the three axes. Under a small rotation about each Euler axis in turn (Figure 1.2), roll and pitch tip the pod’s moment $\mathbf{m}$ off vertical, while yaw (a rotation about $\mathbf{m}$ ’s own direction) leaves it pointing where it started.

body-frame yaw/ torque, any attitude For the single-equivalent-dipole model, the coil-array torque $\boldsymbol{\tau} = \mathbf{m}_{\text{world}} \times \mathbf{B}$ has no component along $\mathbf{m}_{\text{world}}$, for any field and any pod attitude. Torque about the pod’s own dipole axis(yaw) is therefore zero and cannot be produced by any current allocation.

Proof. $\boldsymbol{\tau} = \mathbf{m}_{\text{world}} \times \mathbf{B}$ is orthogonal to $\mathbf{m}_{\text{world}}$ owing to the cross product, so $\hat{\mathbf{m}}_{\text{world}} \cdot \boldsymbol{\tau} = 0$ for every $\mathbf{B}$. A sum of coil contributions is still a field, so this holds for any track layout and any current allocation. This is a property of the $\boldsymbol{\tau} = \mathbf{m} \times \mathbf{B}$ and does not depend on the rotation or composition order.

(saddle yaw-neutrality, level attitude). At $\phi = \theta = 0$, the world-frame moment $\mathbf{m}_{\text{world}} = R_z(\psi) (0, 0, m)^\top = (0, 0, m)^\top$ for every ψ : results in a rotation about z which fixes a vector already lying along z . The potential-energy landscape $U(\mathbf{x}) = -\mathbf{m}_{\text{world}} \cdot \mathbf{B}(\mathbf{x})$ is then independent of ψ , so the location and local geometry of its saddle point are identical for every yaw angle.

Proof. At $\phi = \theta = 0$, $R_x(0) = R_y(0) = I$, so $R(0, 0, \psi) = R_z(\psi)$ and $R_z(\psi) (0, 0, m)^\top = (0, 0, m)^\top$. Because $\mathbf{B}(\mathbf{x})$ depends only on position, $U(\mathbf{x}; 0, 0, \psi)$ is independent of ψ at fixed $\mathbf{x}$.

When leaving a level attitude these characteristics diverge: the full composition $R = R_z(\psi)R_y(\theta)R_x(\phi)$, a nonzero roll or pitch tilts $\mathbf{m}_{\text{body}}$ off $\hat{\mathbf{z}}_b$ before the outer $R_z(\psi)$ acts, so $\mathbf{m}_{\text{world}}$ (and U saddle) do depend on ψ once $\phi \neq 0$ or $\theta \neq 0$; yaw becomes partially reachable under tilt, growing with the angle as it gains a greater component acting on it. So the first is unconditional, with the second

yaw neutrality becoming reachable when not at level-attitude. Roll and pitch, by contrast, are symmetric to *each other* rather than to yaw: both tip $\hat{\mathbf{z}}_b$ off vertical, about x and y respectively, and either one shifts the true saddle-point location.

This asymmetry, with yaw neutral at level attitude, roll and pitch is the central conceptual result of this chapter, and is essential for the construction of a control law that will function to contain the pod at a given point.

1.6 Control mechanism in the simulation

From what has been observed in the literature there are several key features that inform system design.

1.6.1 Actuation Geometry

The coils are arranged in rows and columns in a 2D plane. The current flowing through them produces a field, which can be moderated either through rows or individual coils. Prolusion is envisioned to be performed by rows of coils acting in unison.

At the track centreline a row's coils are laid out mirror-symmetricly, meaning a single shared per-row current cannot produce net lateral force or net roll torque, as a symmetric current distribution has nothing to break the symmetry. This is not a limitation for the independent per-coil currents. The same travelling-wave, excitation is how long-stator maglev propels its vehicles. Transrapid and SCMaglev use a linear synchronous motor built on this principle³¹.

1.6.2 The Active-Control Requirement

Stabilising a saddle-point equilibrium requires a control law, described as: a map from the pod's measured position and orientation to a coil-current command, which is updated fast enough to prevent deviations before they diverge by slipping off the saddle. Its state-dependence determines whether it can track the true instantaneous equilibrium or only a fixed approximation of it. And whether the equilibrium drifts faster than an achievable control loop can respond, with the required sensing, communication, and actuation latency on a distributed track, is an open question for the concept. The questions of the practicality of the centralised coordination framework is outside this dissertation's scope, which is narrowed to aim at the control mechanism, that solely tackles the control laws.

1.7 Active Stabilisation in the Literature

Holding a system that is unstable at an equilibrium without continuous correction is a mature area of control engineering, addressed across a range of physical systems.

1.7.1 Active Magnetic Bearings

An active magnetic bearing (AMB) suspends a rotating shaft with no mechanical contact, using actively-driven electromagnets to hold the shaft's position against the Earnshaw instability. In contrast to NifTI this is a spinning rotor rather than a full moving vehicle. AMBs have been industrially viable for decades³², with their primary purpose being eliminating the mechanical

wear of physical bearings. This is the same driving motivation as NifTI. Additionally, the control problem again has shared features, as an AMB rotor must be stabilised simultaneously against radial (x/y) translation and tilt rotation, with cross-axis and gyroscopic coupling between those channels³³ representing a coupled, multi-axis feedback problem. However, the analogy has a limit, in that where AMB's stabilises a rotor about a mechanically fixed axis, with actuators acting in multiple planes and direction (as seen in Fig.1.3) removing several degrees of freedom by construction; NifTI must stabilise the free-flying body's orientation that is itself moving.

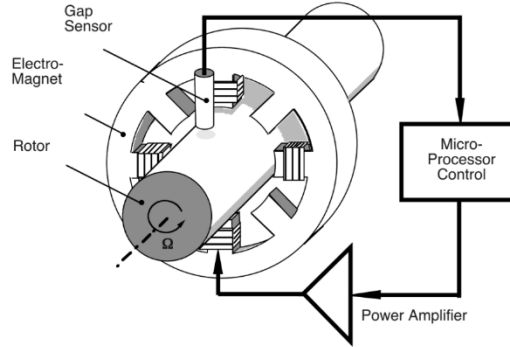


Figure 1.3: Schematic representation of a classic active magnetic bearing³².

1.7.2 Control Strategies for EMS Gap Stabilisation

EMS levitation is inherently unstable and therefore requires feedback control to maintain the levitation gap. The force produced by an electromagnet depends nonlinearly on both current and gap, approximately following $(i/z)^2$. A small reduction in gap increases the attractive force, pulling the vehicle closer to the magnet, while an increase reduces the force. Without feedback, the vehicle therefore moves away from the desired gap.

The standard approach is feedback control, with PID/ PD controllers are commonly used to regulate the gap from its measured error and rate of change^{32;33}. More advanced methods such as feedback linearisation, LQR and model-predictive control are also possible^{34–37}. However, these methods add complexity that is not needed for the initial controller developed here.

A PD controller was therefore chosen as a simple starting point. The existing baseline current assignment method already provides the main steady-state current, so the controller only needs to provide a correction based on the current error and its rate of change. An integral term was not included because it could accumulate error when the available force is limited. The controller is also separated from the current allocation, allowing the control law to be replaced or extended later without changing the rest of the architecture.

Although EMS control is usually discussed in terms of the vertical levitation gap, the same instability occurs in the lateral directions. Active magnetic bearings commonly use feedback control on their radial axes for this reason^{32;33}. The lateral PD controller used in this work therefore applies the same principle to horizontal motion.

The main extension beyond conventional gap control is attitude control. The system developed here also provides control of roll, pitch and yaw, allowing the controller to regulate both the position and orientation of the pod. It is implemented for attitude control in the literature³⁸.

1.8 The Research Gap

Two bodies of work approach what is needed to work on this problem, but neither addresses it directly.

The first is precision magnetic suspension. Active magnetic bearings routinely close the loop on radial position and tilt at once, handling the cross-axis and gyroscopic coupling between those channels^{32;33}, and related systems extend to full six-degree-of-freedom regulation³⁸. What makes this possible is the actuator arrangement: the electromagnets surround the suspended body and act in several directions simultaneously with a high degree of accuracy to understand where the target is. The second is vehicle-scale, track-based EMS. Here the actuators are distributed along a guideway rather than around the vehicle, and the control problem is correspondingly narrower: gap regulation, with lateral guidance where the geometry requires it^{22;23}. Attitude is generally constrained by the mechanical arrangement rather than actively held.

NIfTI sits between the two and inherits the difficulty of both. Like a magnetic bearing it must regulate orientation as well as position, because the location of the saddle point depends on the pod's attitude (§1.5.2). Like a track-based EMS system it must do so from a single planar bed of vertical-axis coils beneath the vehicle. That combination is the gap: what a controller can achieve is bounded not by the control law but by the possible force directions the actuator arrangement can physically produce, and neither literature establishes what those directions are for this geometry. §1.5.2's yaw unreachability is one vast problem that must be overcome, that no choice of gains can remove, because it follows from $\boldsymbol{\tau} = \mathbf{m} \times \mathbf{B}$ rather than from the controller.

Investigating this requires the simulation to be usable as an instrument, and it is not. The inherited implementation is a single 1,861-line script with no regression tests, in which the field model, coil geometry, integrator and current assignment are interleaved. A change to one may silently alter another, and there is no means of establishing that it has not. The gap is therefore two-layered: a **structural** layer, in which the software cannot support controlled experiments; and a **physical** layer, in which the active-control problem is not the position-tracking problem the inherited current schedule treats it as.

1.8.1 Objectives

This dissertation responds to that gap with four objectives:

1. Faithfully reconstruct the inherited simulation as a modular, dependency-ordered code-base, and demonstrate that its behaviour matches the original.
2. Establish which wrench directions the coil array can physically produce under the available pod models, and whether they are sufficient for stabilisation.
3. Add a closed-loop control layer that separates the generation of a corrective wrench from its allocation to coil currents, so that control laws and actuator models can be replaced independently.
4. Determine the range of attitude over which the inherited Euler-angle formulation remains adequate, and provide an alternative beyond it.

Chapters 3 and 4 report against these objectives; Chapter ?? states which were met and which were not.

Chapter 2

Methods & Materials

This chapter describes the material inherited from the preceding work and the methods applied to reconstruct and extend it. The inherited simulation was a single script in which the field model, geometry, rigid-body dynamics and current assignment were mixed together, so no part of it could be changed, tested or replaced in isolation. The method that flows from that starting point, with a set design philosophy: establish what the original does, reproduce it exactly in a structure that separates those concerns, and only then build on it.

Section 2.1 describes the simulation as inherited, its field model and coordinate frames, its rigid-body equations and simplifying assumptions, the numerical scheme, and the current allocation rule that stands in place of a controller. Section 2.2 describes the reconstruction into a modular codebase, together with the provenance tracking, per-package regression tests and runtime parity checks that kept it faithful to the original. Section 2.4, describes the three extensions built on that codebase: a closed-loop control layer that separates wrench generation from current allocation, following the control-allocation architecture used for over-actuated vehicles³⁹; an SO(3) attitude integrator that removes the Euler-angle singularity, following the geometric formulation of Lee *et al.*⁴⁰; and finite-body force models that represent the pod’s magnets at their actual positions rather than as a single point dipole. Section 2.6 sets out the open questions these components raise and the notebook experiments used to answer them.

2.1 The Inherited Simulation

The object of study is the inherited monolithic simulator of magnetic-levitation pod-and-track system. In outline, it works as follows. The user specifies a *track*: rows of cylindrical electro-magnet coils at fixed positions and a *pod*: a rigid body carrying permanent magnets. Each coil, driven by a current, produces a magnetic field; the fields of all coils and permanent magnets interact. That field exerts a force and a torque on the pod’s magnets, which, together with gravity, drive the pod’s rigid-body motion. The simulator then integrates that motion forward in time from the given initial state.

From this core, the simulator produces static maps of the field and of the magnetic potential energy over a region of space; the locations of equilibria and saddle points of that potential; and histories of the pod’s position and attitude under a prescribed set of coil currents (open-loop trajectories). The reconstruction of Section 2.4 preserves all of this behaviour unchanged; the extensions utilise this behaviour.

The underlying physics of the simulation is now explained.

2.1.1 Field model and coordinate frames

Each coil’s axial and radial field components are evaluated from the closed-form elliptic-integral solution for a finite solenoid²⁷, implemented as four `@numba.njit` kernels (`constants`, `elliptic`, `B_rho`, `B_z`) and summed over the coils.

Two coordinate frames are used: a fixed world frame, and a body frame attached to the pod. Attitude is represented by an intrinsic ZYX Euler triple (ϕ, θ, ψ) (roll, pitch, yaw), with $R(\phi, \theta, \psi)$ the body-to-world rotation. The pod’s magnetic moment is fixed in the body frame; with its world-frame value being $\mathbf{m}_{\text{world}} = R \mathbf{m}_{\text{body}}$.

2.1.2 Rigid-body Dynamics

The pods state is represented by a twelve-component vector

$$\mathbf{s} = [x, y, z, \dot{x}, \dot{y}, \dot{z}, \phi, \theta, \psi, \omega_x, \omega_y, \omega_z], \quad (2.1)$$

With positions and velocities in m and m/s, respectively, angles are in radians, and $\boldsymbol{\omega}$ denoting the body angular rates in rad/s. The equations of motion for the simulation are

$$\begin{aligned} \mathbf{F} &= (\mathbf{m}_{\text{world}} \cdot \nabla) \mathbf{B} - mg \hat{\mathbf{z}}, & \ddot{\mathbf{r}} &= \mathbf{F}/m, \\ \boldsymbol{\tau} &= \mathbf{m}_{\text{world}} \times \mathbf{B}, & \dot{\boldsymbol{\omega}} &= I^{-1} \boldsymbol{\tau}, \end{aligned} \quad (2.2)$$

where I is the world-frame inertia tensor, evaluated once at the initial time t_0 . This also comes with two simplifying assumptions, inherited from the code: the gyroscopic term $\boldsymbol{\omega} \times (I\boldsymbol{\omega})$ is omitted, and body angular rates are treated directly as Euler-angle rates, $[\dot{\phi}, \dot{\theta}, \dot{\psi}] = \boldsymbol{\omega}$. This limits attitude accuracy as the angular deviations increase. Additionally, the torque model approximates the pod as a point magnetic dipole located at its centre of mass, giving the standard dipole force $(\mathbf{m}_{\text{world}} \cdot \nabla) \mathbf{B}$ and torque $\mathbf{m}_{\text{world}} \times \mathbf{B}$ ²⁰.

2.1.3 Numerical integration

The choice of solver and its constituent tolerances affects every trajectory-based result. It uses `scipy.integrate.solve_ivp` with the explicit adaptive Runge–Kutta method RK45, with a tolerance 10^{-3} , absolute tolerance 10^{-6} , and maximum step $h_{\text{max}} = 10^{-3}$ s; the solver’s own adaptive nodes are returned. A terminal event halts integration when the pod centre of mass leaves the field domain. These solver tolerances are distinct from the `np.isclose` tolerances used in the regression suite (§2.3.3).

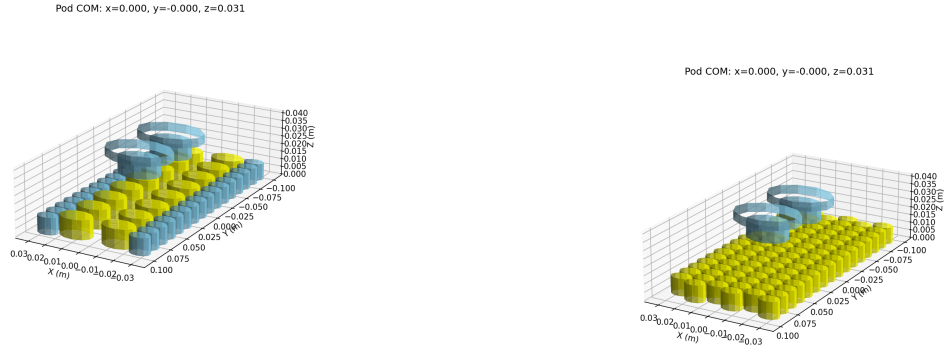
In the open-loop scheme, $\mathbf{B}$ and $\nabla \mathbf{B}$ are read by trilinear `RegularGridInterpolator` from a grid precomputed once per run: at each candidate centre-of-mass position the grid stores $\sum_i \mathbf{B}_{\text{track}}(\mathbf{r}_i)$ which is summed over the pod’s N sample points, with $\nabla \mathbf{B}$ obtained by `numpy.gradient`.

2.1.4 The inherited current schedule

The inherited script has no feedback control. But instead relies on an empirical current schedule (`track/control.py::assign_currents_for_pod`, referred to in prior work as “Eq. 18”). Given the pod’s radial displacement from the starting point using `np.hypot(x, y)` and mass, it assigns a fixed current to each coil row from a fitted functional form. It was originally derived by sweeping coil currents, locating the potential-energy saddle, fitting a quadratic and inverting it. Therefore the schedule is orientation and direction-blind. It cannot sense pitch or roll, cannot distinguish $+x$ from $-x$, and has no settable target.

2.1.5 Inherited geometries and experimental data

Alongside the script, the inherited material comprises the preset track and pod geometries used throughout this work, with coil-row layouts, radii and heights, as well as ferromagnet radii, heights and magnetisations. Experimental data comprised a single bench recording of the magnetic field at eight fixed sensor positions above a small-scale track. The preset geometries are used directly in the simulation to recreate the same physical geometry, and every calibration-relevant check in the regression suite (§2.3.3) is run against them rather than synthetic alternatives. The field recording is used only as a qualitative cross-check, rather than hinging on one specific preset or physical rig. The aim of this was too make the work generalised and not be restricted to any real current setup, but embrace the modular architecture which will allow for future experimentation and iteration.



(a) Classic track configuration, with 4 rows of coils, the outer being ferromagnetic, similar to the pod and the central division being electromagnet coils

(b) Hex track configuration of the pods

Figure 2.1: The two default track presets with pod configuration atop showing pod with configuration two stacks decreasing.

configurations. of these combinations are demonstrated in Figure:2.1, however it is important to note that for all experiments within the system they will be based on the model: 2 stack conic decreasing atop and the hex-track configuration¹⁹

2.2 Reconstructing the Simulation

NIFTI is a long-running, predominantly student-developed project with limited handover between successive implementations. The inherited codebase provided little documentation of its structure and dependencies, concentrated its functionality in one 1,861-line script, and had no regression test suite, so there was no way to tell whether a change to one part of the simulation affected another. The aim of the reconstruction was to establish a modular framework in which the field model, geometry, dynamics and control could be developed, tested and replaced independently, and to separate the simulation from its interactive interface so that components could be driven programmatically for automated experiments. These automated experiments could include parameter sweeps of different poses for the pod or systematically varying the means by which the pod attempts to position itself atop the saddle.

2.3 Principles for the Reconstruction

The refactoring was guided by three methodological principles. First, the internal structure was changed while preserving the observable behaviour of the inherited simulation. Since the pass did not use the full experimental data the original outputs provided the only practical reference for correctness during the reconstruction.

Second, existing behaviour was captured through a regression suite before and throughout the restructuring. These tests were intended to detect unintended changes introduced by the refactor, rather than to establish that the underlying physical model was correct. This distinction allowed the software structure to be changed independently of later questions about model validity.

Finally, the reconstruction was carried out in small, verifiable steps. This made any discrepancy easier to trace to a particular change and was especially important for numerical code, where small implementation differences can propagate into different trajectories. Exact agreement was therefore required during verification (§2.3.3).

The resulting decomposition was based on separating components that could reasonably be changed independently: the field model, track geometry, pod representation, control law and integrator. The effectiveness of these boundaries is assessed in §4.

2.3.1 The extraction process

No architecture for the modularised final system existed at the outset. Therefore the first task was to recover the inherited script’s computational structure, trace how information propagates through the model and identify dependencies between the successive calculations. The geometry of the system determines the field calculation; the field determines the forces and torques on the pod; those pass to the control calculations. Organising the script this way yielded a dependency order (Figure 2.2).

That order then guided extraction. Rather than restructuring in a single pass, functionality was moved out incrementally, one package at a time, in dependency order: each new package could refer only to packages already extracted and verified, which prevents a later-stage component from quietly becoming a dependency of an earlier one. The extraction attempted to copy code verbatim where existing calculations were carried across, and all new functionality was added separately. This also meant that the code was not targetted for optimisation and acceleration as the current state did not necessitate it as the largest block was the starting parameters of the simulation and using the UI.

For the refactor, however, every step was small enough that a discrepancy could be attributed to a single move rather than to many simultaneous edits, which made verification alongside the refactor possible (§2.3.3) rather than only at the end. Second, changes caused by modularisation can be told apart from changes caused by later model development, because the two were never mixed in one commit. This incremental, behaviour-preserving approach is the standard one for restructuring an existing system without a safety net of tests^{41;42}.

2.3.2 The resulting package structure

Extraction produced seven packages, each depending only on those below it in the order of Figure 2.2. Their responsibilities, in dependency order:

Table 2.1: Structure of the reconstructed codebase. The packages separate the original monolithic script into geometry, physics, track definition, control, data handling, simulation and plotting. The `config` and `naming` modules sit alongside these packages, while the tests provide regression checks for the reconstructed implementation.

Package	Responsibility and constituent modules	Origin
<code>config</code> (1 module)	<code>config.py</code> holds the physical constants, output paths and default simulation settings. These values were previously scattered through the original script. The new <code>SimConfig</code> dataclass keeps them together.	Values ported; <code>SimConfig</code> new.
<code>naming</code> (1 module)	<code>naming.py</code> handles run names and output filenames. The <code>RunTag</code> dataclass replaces the mutable filename globals used by the original.	New.
<code>geometry</code> (5 modules + __init__)	This package contains the geometry and rigid-body calculations, without any field calculations. <code>shapes.py</code> defines the magnet types. <code>collection.py</code> groups magnets into rigid bodies and calculates their mass, volume and inertia. <code>sampling.py</code> creates the dipole point clouds used by the energy sweeps. <code>transforms.py</code> handles Euler-angle rotations, while <code>quaternion.py</code> provides the quaternion and SO(3) operations used by the attitude integrator.	Core geometry modules ported; <code>quaternion.py</code> and body-frame inertia new.
<code>physics</code> (5 modules)	This package contains the field and force calculations. <code>field.py</code> provides the finite-cylinder field solution. <code>field_query.py</code> applies it to magnets and complete tracks. <code>dipole.py</code> contains the point-dipole and multi-dipole pod models. The multi-dipole model also captures the lever-arm contribution to torque. <code>ideal_field.py</code> provides the analytical comparison fields. <code>plant_linearization.py</code> calculates the position-stiffness Jacobian used when checking the controller gains.	Core field modules ported; <code>dipole.py</code> and <code>linearise.py</code> new.
<code>track</code> (2 modules)	This package builds the track and applies the inherited current schedule. <code>geometry.py</code> converts preset rows into track magnets. <code>control.py</code> calculates the baseline current for each coil from the pod position, ring and scaling factors. The duplicated schedule logic in the original was reduced to one implementation.	Schedule ported and de-duplicated; geometry new.
<code>control</code> (6 modules + __init__)	This package contains the new closed-loop control system. <code>base.py</code> defines the controller interface and wrench command. <code>xy_controller.py</code> and <code>z_controller.py</code> provide translational PD control. <code>attitude_controller.py</code> provides the SO(3) attitude controller for roll and pitch. <code>combined_controller.py</code> combines these commands. <code>allocation.py</code> converts the requested wrench into current corrections on top of the baseline schedule. It supports both point and multi-dipole pod models.	Entirely new.

Continued on next page

Package	Responsibility and constituent modules	Origin
<code>data_io</code> (4 modules)	seperates file handling from the simulation. <code>presets.py</code> loads track and pod geometries. <code>field_data.py</code> handles field and energy grids. <code>results.py</code> saves simulation results. <code>sensor_data.py</code> loads the experimental sensor data. The reconstruction also removes the baked in CLI work.	All four ported; prompting removed and one unit bug fixed.
<code>simulation</code> (6 modules)	This package runs the full simulations. <code>dynamics.py</code> contains the rigid-body equations and both Euler-angle and quaternion integrators. <code>closed_loop.py</code> runs the original Euler-based closed-loop path. <code>closed_loop_so3.py</code> provides the new quaternion-based path with lateral, vertical and attitude control. <code>calibration.py</code> finds the current scaling needed for levitation. <code>potential.py</code> runs the potential-energy sweeps, and <code>sensor_check.py</code> performs the field validation.	Original Euler path, potential and sensor checks ported; the SO(3) path, closed-loop drivers and calibration new.
<code>plotting</code> (3 modules)	This package is responsible only for visualisation. <code>geometry_plots.py</code> shows the track and pod. <code>heatmaps.py</code> produces the potential-energy maps. <code>trajectory.py</code> plots position and orientation from the simulation output.	Existing plotting code ported and separated.
<code>tests</code> (4 test files + 2 support)	The test suite checks that the reconstructed modules behave consistently. <code>test_core.py</code> covers the basic configuration, geometry and track layers. <code>test_physics.py</code> checks the field and dipole calculations. <code>test_control.py</code> covers the controllers and current allocation. <code>test_simulation.py</code> checks the dynamics, calibration and simulation paths. <code>run_all.py</code> executes the tests in dependency order, with <code>_helpers.py</code> providing shared setup.	Entirely new.

`naming.py` and `config.py` sit outside of the chain and are responsible for replacing six mutable global states from the monolith. These are exported to the `save` and `plot` functions, allowing for two runs to coexist in one process without their outputs colliding. Whereas `Config` allows for changing of experimental parameters without entering the core of the physics simulations.

2.3.3 Verifying the reconstruction

Two mechanisms ran continuously during the reconstruction, to ensure validation with a third added as a final demonstration of parity.

Provenance tracking. Every function moved during the refactor is tagged, at the moment it is moved, with its original source line range in `baseline_code.py` and a status (unchanged / modified / new), recorded in `PROVENANCE.md`. Tracking provenance during the move, rather than reconstructing it afterwards, ensures that the code being identical to the original was directly checkable on a function by function basis.

Regression suite. Tests are an essential part of software engineering, and especially important when building a simulation this complex and layered. The tests are written as plain Python scripts, in four packages, collected in `tests/` and run in dependency order by `run_all.py`, which stops at the first failure. A framework such as `pytest` was deliberately not adopted, as the goal

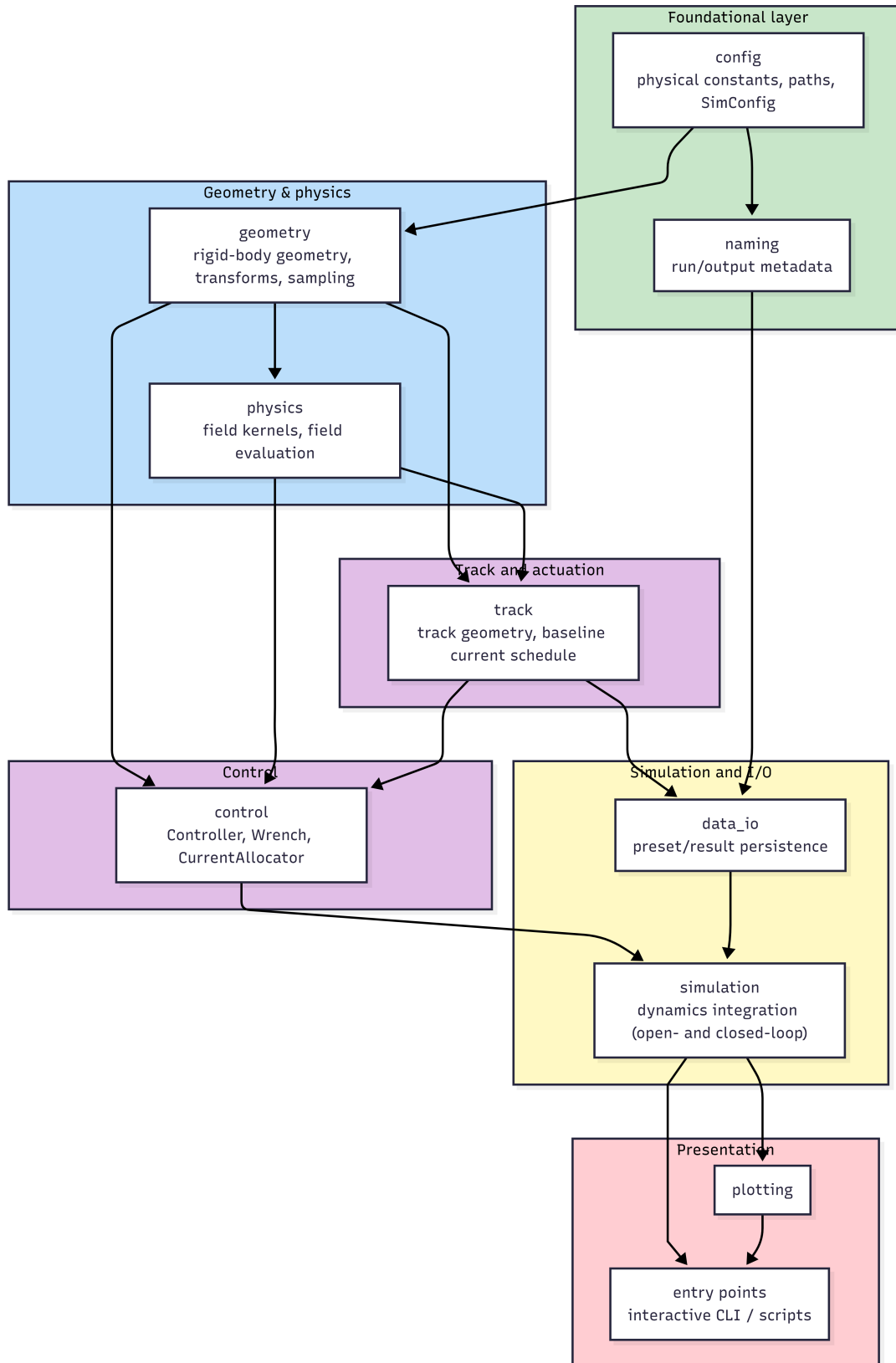


Figure 2.2: The reconstructed codebase, in dependency order: seven packages and the two standalone modules on naming and config. Each package depends only on those above it; an arrow runs from a dependency to the package that uses it. This is the structure the extensions build on.

was to confirm that each newly extracted package still reproduced the behaviour established in the packages beneath it, which a direct per-package pass/fail check is sufficient for. Because each package carried this coverage from the point of extraction, a few small bugs were caught and localised to a single package as they were introduced, not as an unexplained discrepancy in a full run closed-loop output.

Runtime parity against the monolith. The provenance and regression tests establish the self-consistency; neither, on its own, test the output agrees with `baseline_code.py` at runtime. This motivated the construction of two executable checks, first a run of six copied field and rotation functions through both code paths, and an end-to-end check that runs the monolith’s own `plot_movement` integrator against the refactor. This is omitted from the test suite so as to make it non essential for the original script to be in the directory.

At this point the simulation had been fully reconstructed, with an accompanying test suite validating its correctness.

2.3.4 A stopping Point

With the completion of the modularisation of the code the primary objective of the research project had been reached. The code had isolated sections from outside influence. The modularisation now enabled what was promised at the outset in that it enabled extensibility and investigation of the systems constituent parts.

2.4 Closed-loop control

The first section that required investigation was building feedback into the system. As as inherited the simulation has no feedback. Every coil current comes from a fixed schedule that knows only the pod’s radial distance $r = \text{hypot}(x, y)$ from the track centreline, the baseline equation was produced in previous work within the project¹⁹:

$$I_i(x, y) = \frac{k}{1000} \left[1.8 + 0.032 (2.5 \rho_i + 1221.08 r)^2 + 39.34 r \right]$$

where ρ_i is coil ring i ’s normalised radius and k rescales the fit by the simulated pod’s mass and coil count relative to the reference pod it was fitted to ($m_{\text{ref}} = 0.1459$ kg, $N_{\text{ref}} = 78$). The schedule is blind to the sign of x, y, z , and pod attitude, and it has no target; the coefficients 1.8, 0.032, 1221.08, 39.34 and the 1/1000 factor is manual empirical fit form the previous work[?].

The new control layer does not replace this schedule, it still runs unchanged at every step. This is because this is the only part of the system that is fundamentally calibrated to work against real experimental data. A controller computes the corrective force and torque, and an allocator turns that into a per-coil current added on top, so at zero correction the simulation behaves exactly as before.

Splitting how much force to apply from how the coils apply was motivated by standard construction of control allocation architecture used for overactuated vehicles, including submarines and aircraft³⁹. The controller and allocator share one object, a force–torque pair (a `Wrench`⁴³), so either side can be swapped or stacked without the other knowing about it (Fig. 2.3). The allocator itself is a single linear solve against an exact per-ampere current-to-wrench map (Appendix A), with no extra field evaluations per step, which is a strong efficiency gain.

2.4.1 Control Law Architecture

Three laws run at once and their wrenches are summed (Fig. 2.3). Each is a proportional-derivative (PD) law (the typical form for active magnetic suspension)^{32;33}, with a spring term set by how far a coordinate is from its target, plus a damper term based on how fast it is moving. What differs law to law is which coordinate it acts on and why the associated coordinate was unstable in the first place.

2.4.2 Lateral control

The original control system only measures distance from centre ($\text{hypot}(x, y)$) with no target setpoint, so it cannot hold the pod at a specific lateral position. The Results chapter shows its lateral force pushes the pod *away* from the operating point (anti-restoring), and no adjustment to the control map can fix this.

From the literature, a closed-loop feedback controller was copied to maintain lateral position.

Implement a standard PD controller: the corrective force depends on both the current position error and how fast the pod is moving sideways,

$$\mathbf{F}_{xy} = -k_p \mathbf{e}_{xy} - k_d \dot{\mathbf{r}}_{xy},$$

where $\mathbf{e}_{xy}$ is the offset from the target position, $\dot{\mathbf{r}}_{xy}$ the lateral velocity, k_p the proportional (spring) gain multiplying the error, and k_d the derivative (damper) gain multiplying the velocity. The output force is also capped at a maximum value. This law controls only lateral forces; vertical thrust and rotation are handled separately by the other control loops.

2.4.3 Vertical control

The system's lift scale is unreliable (§2.1.4): the force the coils produce does not consistently result in levitation for the toy example. The vertical PD controller acts on height alone with a force limit, and is otherwise identical in construction to the lateral law, but is tuned far stiffer. The reason is not that height matters more in itself, but that the field the pod sits in is a strong function of height: a small error in z changes not only vertical force but the lateral force and torque computed at that pose, so holding z tightly is what keeps the other two channels well-characterised rather than evaluated at a drifting operating point. A side effect of that stiffness is that it lets the lateral demonstration be measured against a held height rather than a collapsing one: without active vertical control the pod free-falls, and a lateral error taken mid-fall is hard to separate from the pod simply coasting out of the field. The lateral run in §3.7 is therefore reported with this loop both open and engaged, and only the height-held version is used to support a claim about lateral authority.

2.4.4 Attitude control

Nothing in the inherited script regulates orientation at all, both roll and pitch evolve however the rigid-body dynamics take them. A PD law written directly on Euler angles would inherit the issues that result as of the singularities that occur as pitch changes around $\pm 90^\circ$ (§2.5.1), so the law instead works on rotation matrices⁴⁰,

$$\boldsymbol{\tau} = -k_R \mathbf{e}_R - k_\omega \boldsymbol{\omega},$$

Law	Gain	Units	Role
Lateral PD	$k_p \approx 180$	N/m	force per metre of x, y error
	$k_d \approx 5$	N s/m	force per unit lateral speed
	$F_{\max} \approx 3.2$	N	cap on the horizontal force
Vertical PD	$k_p \approx 5000$	N/m	force per metre of z error
	$k_d \approx 20$	N s/m	force per unit vertical speed
	$F_{\max} \approx 1.0$	N	cap on the vertical force
Attitude PD	$k_R \approx 0.60$	N m	torque per unit orientation error
	$k_\omega \approx 1.7 \times 10^{-3}$	N m s	torque per unit angular rate
	$\tau_{\max} \approx 0.088$	N m	cap on the torque

Table 2.2: One illustrative gain set (`LateralPDController`, `ZPDController`, `GeometricAttitudePDController` as found in `control/`. These values are purely representative. The vertical loop is deliberately stiff for the height-sensitivity reason given in §2.4. Identifying gains against rig data is future work (§??).

with $\mathbf{e}_R$ a measure of how far the pod is from the target orientation (near level, just the roll and pitch errors) and $\boldsymbol{\omega}$ the angular velocity. Its yaw component is forced to zero before allocation, owing to yaw unreachability.

2.4.5 Gains

All three laws’ gains are approximate rather than fixed constants. Attempts were made at dynamic selection of the parameters via model optimisation; this was unsuccessful and not grounded in the science of the system. Instead they have been set as parameters that let the system apply enough force to attempt to stabilise itself. The values are shown in Table 2.2.

The table’s vertical restoring force is far higher than any other entry, for the reason given in §2.4: the field is height-sensitive enough that a small z error corrupts the other two channels, not because holding height matters more in itself.

The values currently assigned are an illustrative set, not a tuning result, and the two large gains in the table are large for different reasons. The vertical k_p is large because of the height-sensitivity argument above. The lateral k_p is large for an unrelated reason: it is a stiffness in N/m that must first cancel the schedule’s anti-restoring stiffness before it can add any net restoring force of its own; what is left sets an ordinary few-hertz position loop for a sub-kilogram pod. This is unrelated to the current-scale factor used for levitation, which acts only on the baseline F_z , never on the controller.

2.4.6 Resulting Flow of the Simulation

The outer loop advances the run in a fixed control interval $\Delta t_c = 5$ ms, chosen well within the delay margin reported in the literature for electromagnetic-suspension gap-feedback loops before dynamic response begins to degrade⁴⁴. At each boundary the baseline schedule is re-trimmed to the pod’s current position, the controllers return their wrenches, and the allocator’s resulting current correction is held constant while the interval is integrated with the RK45 settings of §2.1.3. The baseline schedule does not by itself lift the pod for the presets studied, so closed-loop runs wrap it in a per-pose current-scale calibration (`simulation/calibration.py`) that sets a defined start height.

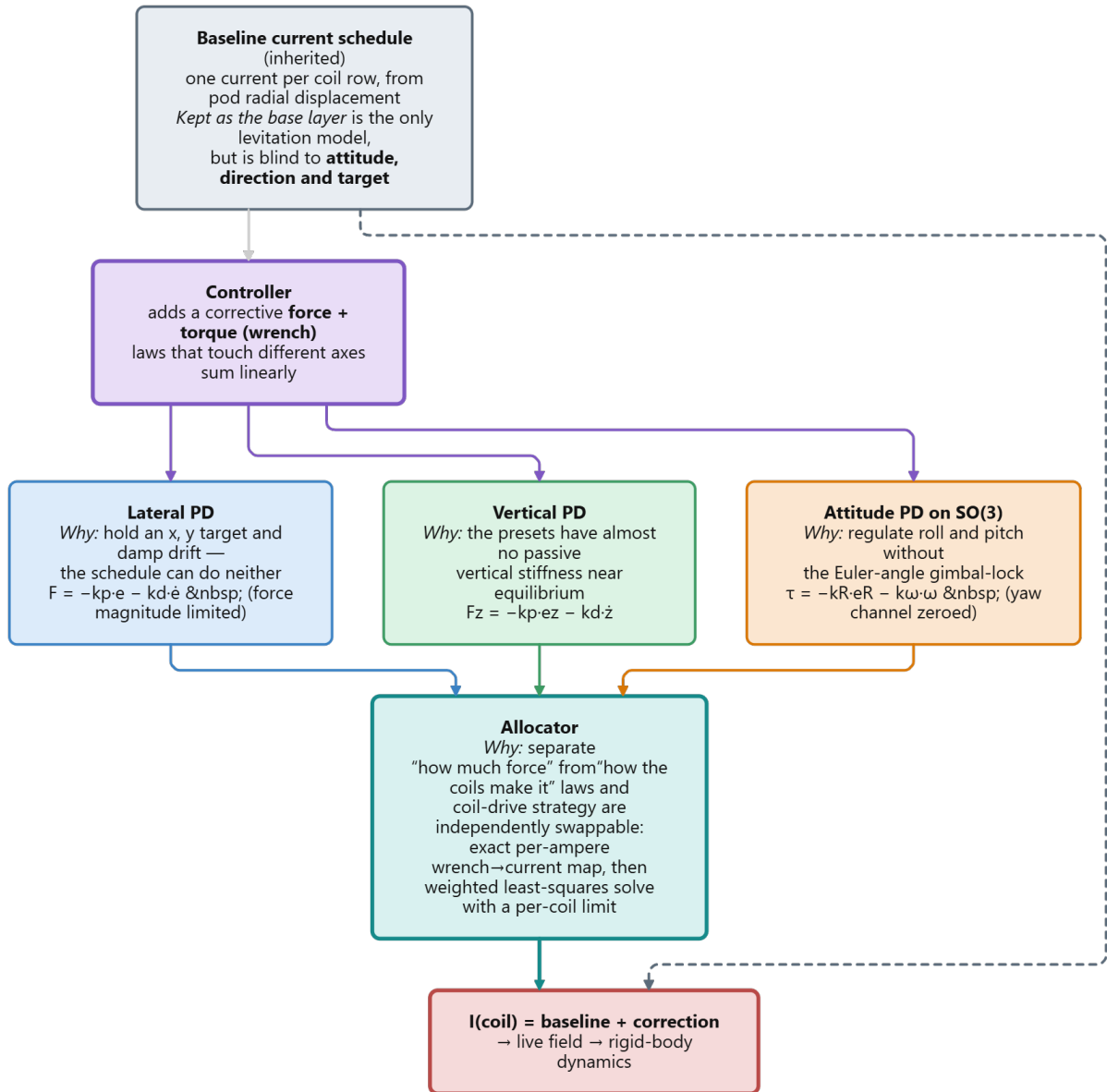


Figure 2.3: The closed-loop control layer. The inherited schedule stays live as the trim layer; each control law adds a wrench correction, the allocator converts the sum to a per-coil current, and that current is added on top of the schedule.

2.5 Additional Improvements

2.5.1 Attitude representation on SO(3)

The inherited integrator stores attitude as ZYX Euler angles, which are singular at pitch $\pm 90^\circ$. A second integrator was added (`_rigid_body_derivatives_so3`) that stores the pods attitude as a unit quaternion, integrates the full rigid-body equation, and renormalises the quaternion at each control step. It can represent motions the Euler version cannot: torque-free tumbles, and any path through the $\pm 90^\circ$ band. Real prototypes would not be flown at such attitudes, but representing them is needed before the model can go beyond near-level motion. *Open question.* Over what range of attitude excursion do the two representations agree, and how does the Euler-angle path fail beyond it? Examined in notebook 05.

2.5.2 Finite-body force models

The basic torque law treats the pod as a single point dipole. Two more detailed models are available: `multi_dipole_force_torque` treats the pod's magnets as separate bodies with their own lever arms, and a finer model splits each magnet into a cloud of sample points, reusing the point-sampling already in `geometry`. The finer model is the reference the other two are measured against. *Open question.* How much does finite-body geometry change the force and torque on the pod, at zero tilt and as the pod tilts? Examined in notebook 04.

2.5.3 Multi Point Dipole

Another key assumption of the work is that the full pod with its multiple sets of magnet stacks can be distilled into one magnetic dipole point at the centroid to calculate the net force on the pod. This is computationally efficient but lacking in depth but prevents forces from acting through lever arms.

This was discovered throughout the process of doing the modularisation, and prompted the construction of the multi-dipole model which represents each pod magnet at its actual position. The force and local dipole torque are calculated for each magnet, with the additional lever-arm torque

$$\boldsymbol{\tau}_i = (\mathbf{r}_i - \mathbf{r}_{\text{COM}}) \times \mathbf{F}_i.$$

Summing these contributions gives the total wrench about the centre of mass. Separated magnets can therefore produce yaw torque, giving the allocator access to all six degrees of freedom.

The

2.6 Open Questions and the Notebook Experiments

The additional models and controllers are kept separate from the main simulation and tested through four dedicated notebooks. This allows individual components to be changed and tested without altering the main execution pipeline.

The notebooks investigate four main questions:

1. **Does the modular code reproduce the original simulator?** (§2.3.3.) *Notebook 01* compares the original and modular implementations using the same inputs. It also examines the potential-energy landscape around the nominal equilibrium to identify the stable and unstable directions.
2. **What forces and torques can the actuators produce?** (§2.4, §2.5.2.) *Notebook 02* compares the point- and multi-dipole pod models and different actuator groupings. It determines which wrench directions are controllable and measures the difference between the two pod models.
3. **Can feedback control improve the inherited baseline?** (§2.4.) *Notebook 03* tests the lateral PD controller and examines its effect on lateral position, height and attitude. It also investigates the vertical force available from the inherited current schedule.
4. **When does the Euler-angle attitude model break down?** (§2.5.1.) *Notebook 04* compares the Euler-angle and $SO(3)$ integrators during a torque-free tumble, establishing the range over which the simpler Euler formulation remains accurate.

The notebooks provide a separate testing environment for the additional models and controllers, while the main simulation remains unchanged. The results of these experiments are presented in the following chapter.

Chapter 3

Results

This work establishes that the original simulation was successfully modularised without changing its behaviour. This provided a framework for extending the code with improved attitude integration, actuator models, and feedback control. The new components demonstrate capabilities that were not present in the inherited simulator, but they are not presented as a tuned or experimentally validated levitation system. A problem of the original code was that the inherited current allocation method for driving the coils (after called the current schedule) does not provide a stable levitation equilibrium, which limits what can be demonstrated by the new controllers. The results therefore separate three questions: whether the original simulation was preserved, what additional capabilities the new architecture provides, and which limitations remain in the original physical and control model.

3.1 Verifying the Refactor

The first requirement was to establish that the original monolithic simulation had been successfully decomposed into approximately 35 modules across seven packages without changing its underlying behaviour. The port was tracked in `PROVENANCE.md`, recording where original features were mapped into the new architecture. Directly ported components were checked using both low-level parity tests and an end-to-end trajectory comparison, while newly developed components were tested separately.

3.1.1 End-to-end validation

The strongest evidence that modularisation preserved the original simulation is the end-to-end parity test performed in Notebook 01. The original monolithic implementation and new modular architecture were run using the same field grid and initial state, with only the surrounding code path differing (Figure 3.1).

This provides a stronger test than comparing individual formulas because it tests whether the restructured components still interact correctly within the complete simulation. The comparison directly tests the original dynamics integrator, `plot_movement` in `baseline_code.py`, against `simulation/dynamics.py::run_dynamics`.

For this illustrative test, the pod was released from rest at $x = y = 0$ and $z = 0.04536$ m on the Hex Track preset. Over the 0.1 s simulation window, all twelve state components agreed exactly (six shown), with a difference of 0.0 throughout. This agreement persisted during a nonlinear instability, with pitch growing from 0 to approximately -1.75 rad. The test was

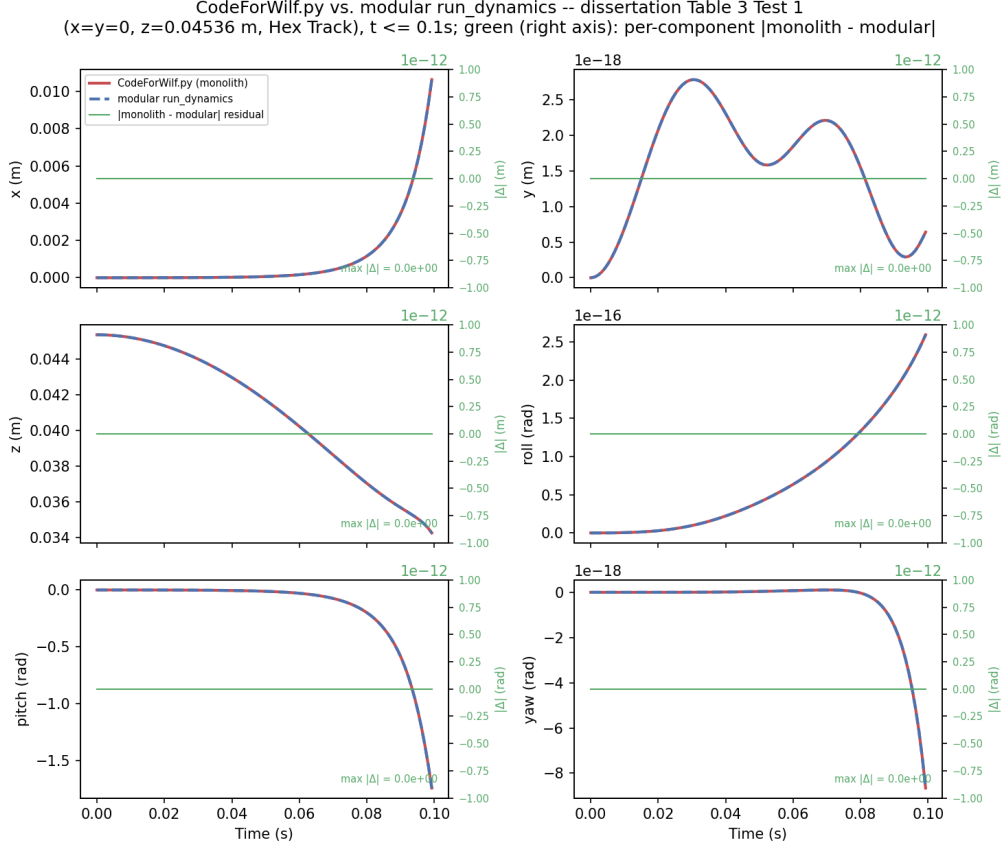


Figure 3.1: `baseline_code.py`'s dynamics integrator compared with the modular `run_dynamics`, using the same field grid, `Collection`, and initial state. The two implementations agree exactly through a developing instability.

selected because the developing instability provides a more demanding parity check than a near-equilibrium trajectory.

Comparison of Numerical Kernels

Runtime comparison of the copied kernels is also necessary. Previous claims that the old and new code were numerically identical relied on manual, line-by-line comparison in `PROVENANCE.md`; the refactor provides a stronger check by running both implementations and comparing their outputs. Six key elliptic-related functions were tested, as shown in Table 3.1.

Outcome of Validation

The tests confirm that the copied functions retain their macroscopic numerical behaviour after being moved into the new code structure. It was able to fully replicate an unstable rapidly diverging simulation example. This end-to-end sweep and the individual-kernel level tests demonstrate the full simulation behaving as expected after the refactor.

With the code reconstructed it allowed for a fundamental shift in the use of the code. It has gone from a single CLI script to a full suite of packages that can be run directly from scripts by importing the library. This feature of the code will be the primary means of testing as batch parameter sweeps can be enabled as well easy configuration of the experimental parameters,

function	role	max $ \Delta $
<code>constants</code>	@numba.njit field kernel	0.0
<code>elliptic</code>	@numba.njit field kernel	0.0
<code>B_rho</code>	@numba.njit field kernel	0.0
<code>B_z</code>	@numba.njit field kernel	0.0
<code>closed_cyl</code>	closed-form multi-magnet field, on a real “Two Stacks” pod	0.0
<code>rotation_matrix_from_euler</code>	geometry	0.0

Table 3.1: Elliptic-related functions and their roles.

either in the config file directly or in a notebook.

3.2 Test Suite

Another key deliverable was a full test suite to complement the completed code. These were subdivided into 4 packages with a single run file at the top. These were primarily smoke tests and smaller tests written for small issues that were interacted with during the run. This was essential for the project as it was what was able to depict that the function that assigned coil currents (`current_allocator`) was orders of magnitude too small, motivating the creation of `calibrate_current`.

3.3 Accuracy of the Attitude Integrator

With the original behaviour established, the modular structure could also be used to introduce a new attitude representation. `simulation/dynamics.py` contains two integration methods: the original Euler-angle method, `_rigid_body_derivatives`, and the newly developed SO(3)/quaternion path, `_rigid_body_derivatives_so3`.

The comparison uses a torque-free rigid-body tumble with $\mathbf{B} = 0$, $\mathbf{gradB} = 0$, and zero dipole moment. In this case angular momentum and rotational kinetic energy should be conserved, providing a direct test of the attitude representation rather than of the magnetic model.

3.3.1 Observations in Attitude

The tumble hits the $\pm 90^\circ$ pitch band several times. These rebounds back in the opposite direction cause the ZYX Euler representation to exhibit the roll and yaw jumps shown in Figure 3.2a. The SO(3) representation passes through the same rotations somewhat smoothly after the initial oscillatory period, while angular momentum remains close to its initial value, with a relative error of approximately 10^{-13} . The Euler-angle system instead develops a large momentum error after the first axis switch.

For smaller rotations, the two representations agree closely. Across a sweep of increasing total rotation, the discrepancy remains below a milliradian in the near-level region (10 $\circ$) relevant to the levitation work, but grows rapidly at larger excursions (Figure 3.3).

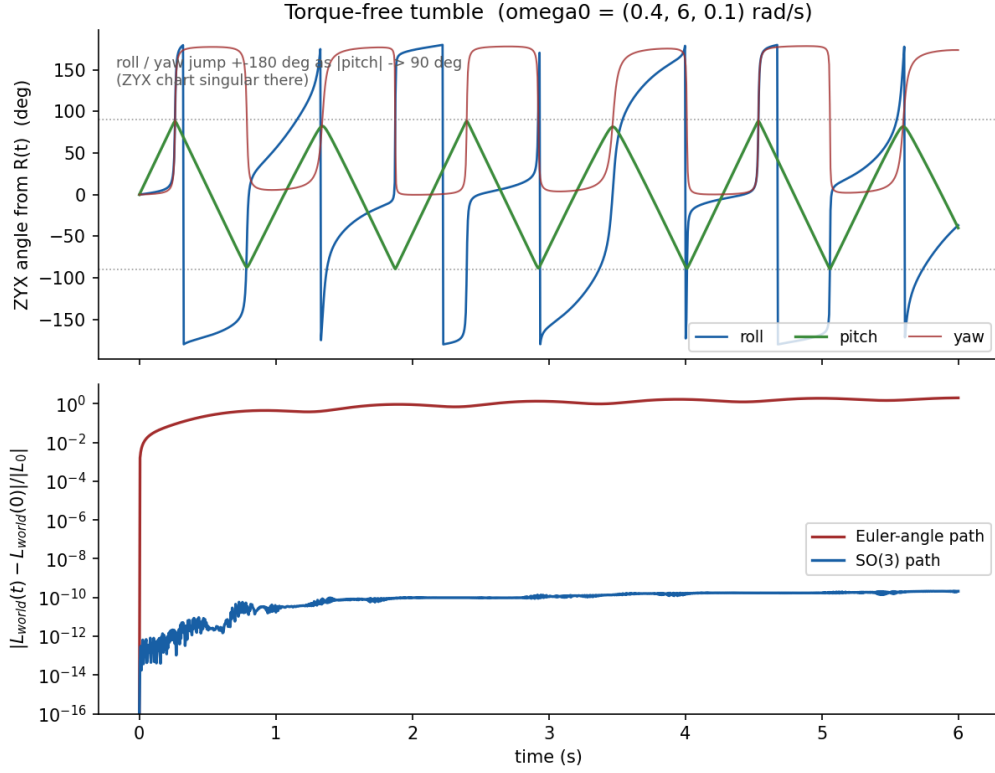


Figure 3.2: Torque-free tumble integrated using both attitude representations. Recovered ZYX angles show the coordinate singularity as the body enters the $\pm 90^\circ$ pitch band, while the quaternion state passes through smoothly. Angular momentum provides the frame-independent check: the SO(3) path remains near 10^{-13} of its starting value, whereas the Euler-angle path loses conservation.

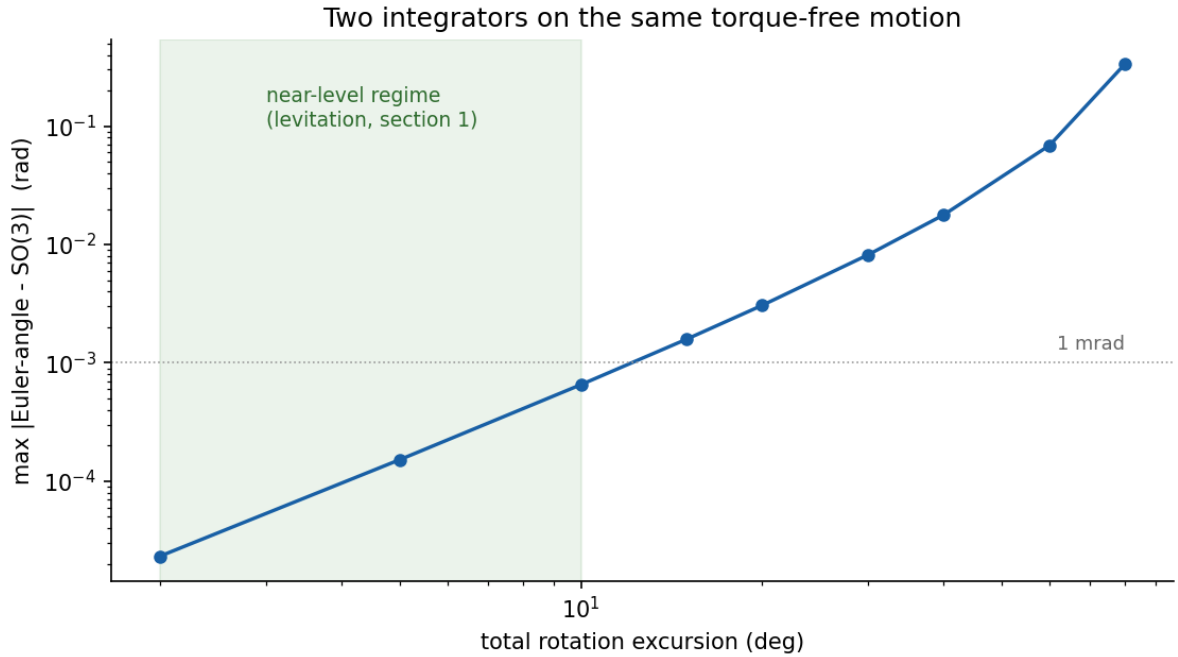


Figure 3.3: Largest angle-wise discrepancy between the two integrators against total rotation excursion. In the near-level region relevant to levitation (shaded), the discrepancy remains below a milliradian; larger excursions increasingly favour the SO(3) representation.

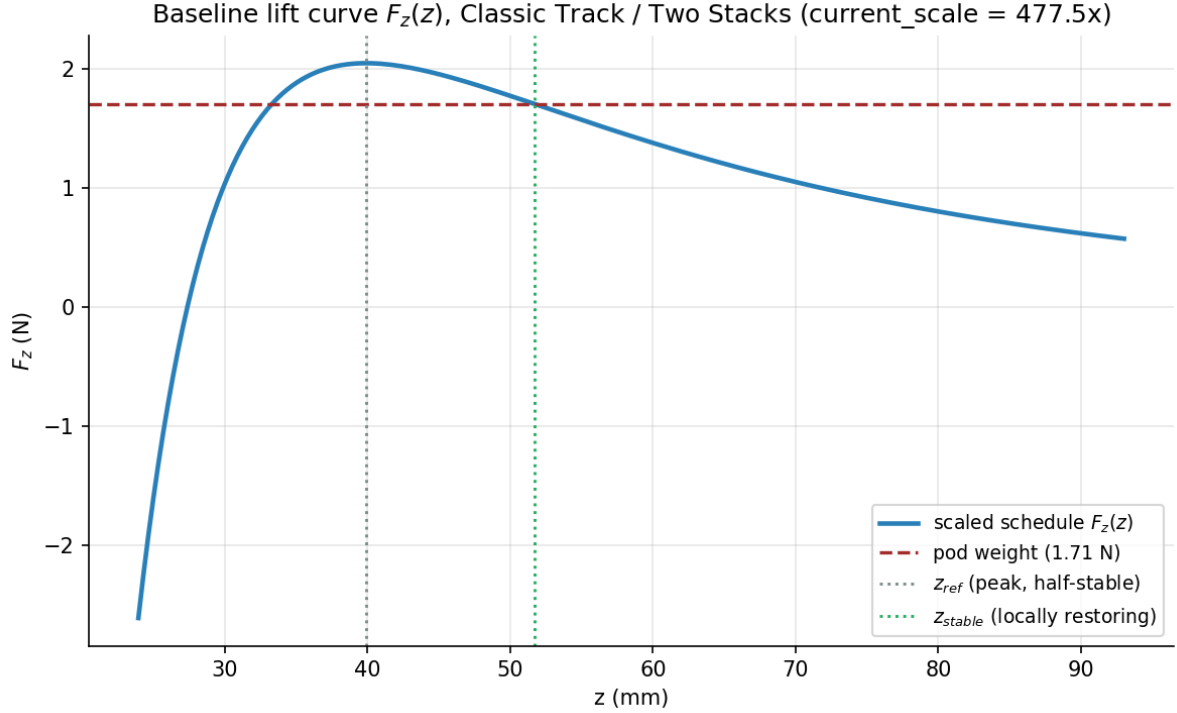


Figure 3.4: Baseline lift curve after scaling. The raw peak is hundreds of times below the pod weight before scaling and provides only limited local vertical stability.

3.3.2 Attitude Conclusions

The $SO(3)$ representation therefore provides a more robust representation when large attitude changes occur, while the original representation remains close to it for the small-angle regime.

3.4 Deficiencies in Original Control Process

Before assessing feedback control, the inherited current schedule must be established as a baseline against which the new control components can be interpreted. The original schedule, implemented in `track/control.py::assign_currents_for_pod`, does not achieve levitation under any tested track/pod combination in `simulation/calibration.py`.

The vertical force curve provides the first limitation. The raw peak force is hundreds of times below the pod weight (calculated as $477.5x$), this is a clear calibration error that has previously been made with the current assignment per coil.

A calibration was made which can find a current scale that balances the pod at a selected height, but this does not mean that the schedule naturally holds that equilibrium. The calibration finds the schedule's peak-lift height z_{ref} and a locally restoring z_{stable} just beyond it, together with a `current_scale` that raises the available force above the pod weight. The current is repeatedly calibrated to balance weight at the current pose.

The lateral behaviour presents a more fundamental stability problem. At the operating point, the linearised environment stiffness is approximately $+38.8$ N/m, while a direct finite-difference calculation gives approximately $+42$ N/m. The agreement in both sign and magnitude confirms that the centre point is anti-restoring in x .

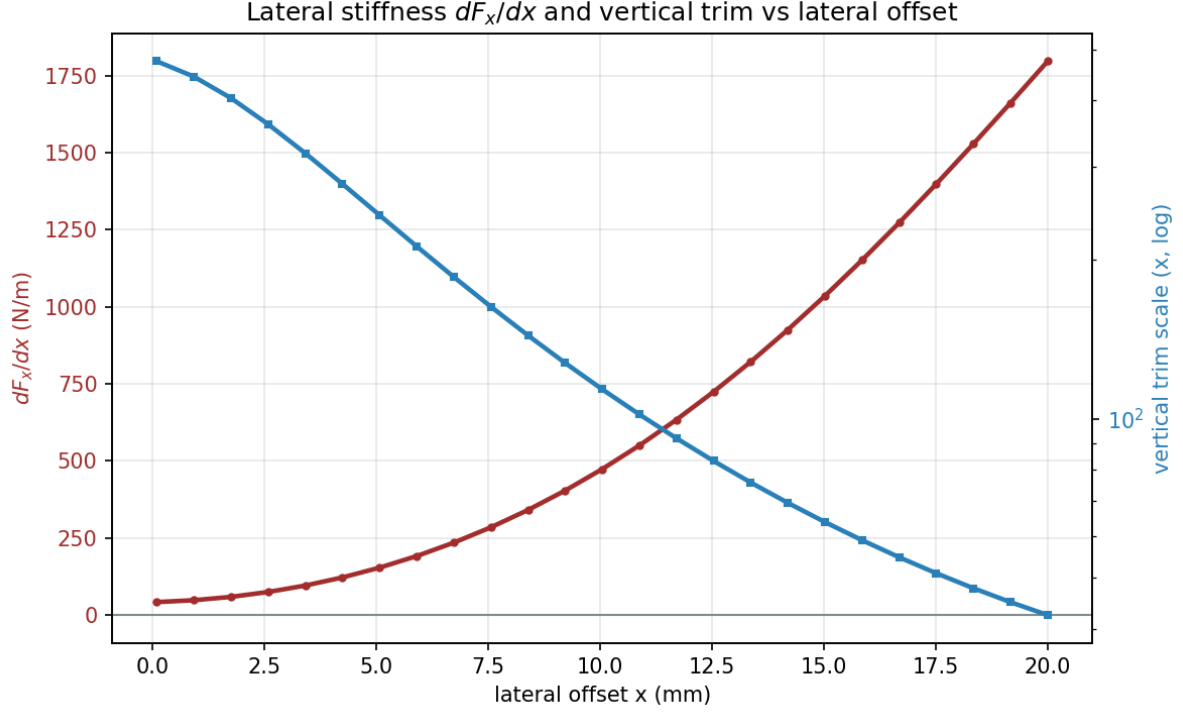


Figure 3.5: Off-centre behaviour of the original schedule. The lateral stiffness dF_x/dx is anti-restoring at the operating point and increases rapidly with displacement, while the vertical trim `current_scale` changes several-fold.

The stiffness increases strongly away from the centre, reaching approximately +1800 N/m at a 20 mm offset. The corresponding vertical trim also changes substantially: `local_current_scale`, the scale required to maintain $F_z = \text{weight}$, varies greatly over the same few millimetres. This behaviour is produced by the original schedule's dependence on $\text{hypot}(x, y)$.

The original therefore provides a narrow central calibration rather than a stable operating region. Figure 3.6 depicts this clearly, as the X component force increases in accordance with the shift from (0,0), All while its vertical force must be actively recalibrated to maintain height.

These results establish the central limitation of the inherited model: it can be calibrated to balance vertical force at the centre, but it does not provide a stable levitation equilibrium. The subsequent controller tests must therefore be interpreted as tests of control authority and software integration rather than demonstrations of stable levitation.

3.5 Identifying Deficiencies in Original Model Stability

The levitation method leads to an unstable equilibrium. The equilibrium in Fig. 3.7 is a saddle point: it is stable in z near the selected operating point but unstable along x . The force field demonstrates this, as the arrows converge toward the operating point along z and diverge from it along x .

With fixed magnetic moment, the potential is

$$U = Mgz - \mathbf{m} \cdot \mathbf{B}. \quad (3.1)$$

In a current-free region, the magnetic contribution is harmonic, therefore its Hessian has zero

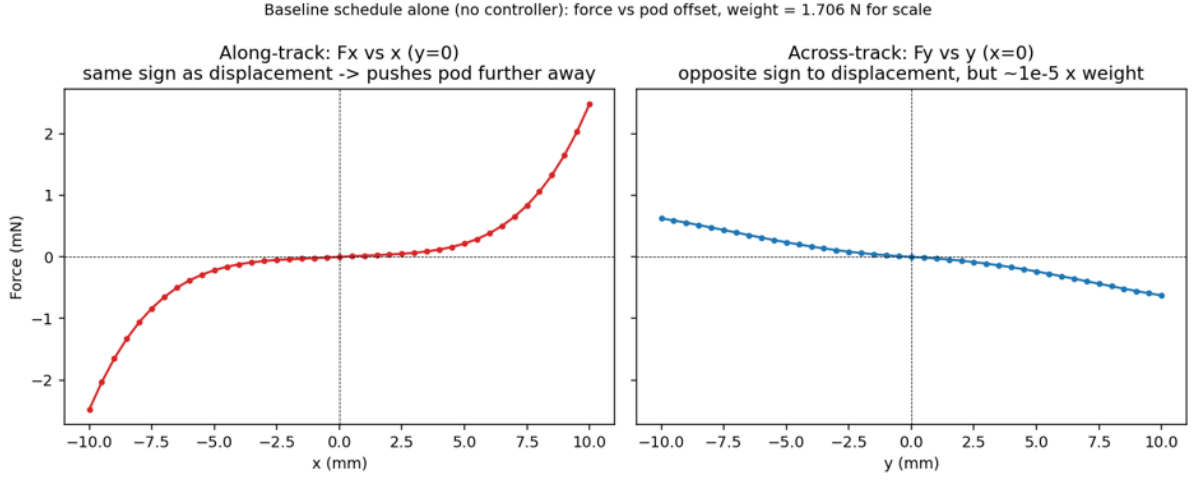


Figure 3.6: Force against distance from the centre produced by the original control-law scheme.

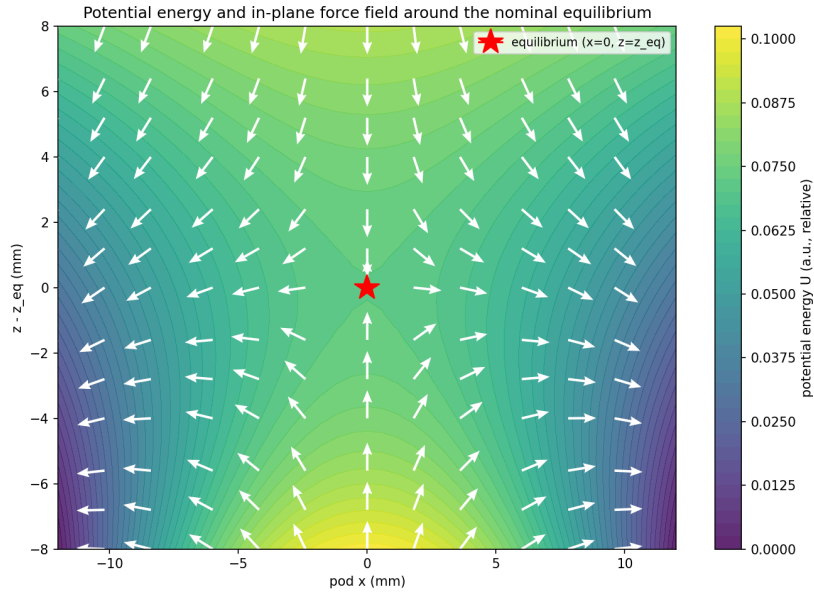


Figure 3.7: Potential and in-plane force field around the nominal equilibrium (notebook 01). Arrows are the normalised numerical gradient $-\nabla U$; the star marks $(x = 0, z = z_{eq})$. The field is restoring along z and anti-restoring along x : a stationary point of a harmonic potential is a saddle, stable in at most two of the three translational directions.

$\tau = \mathbf{m} \times \mathbf{B}$ with $\mathbf{m}$ fixed vertical, $\mathbf{B}$ swept over the sphere

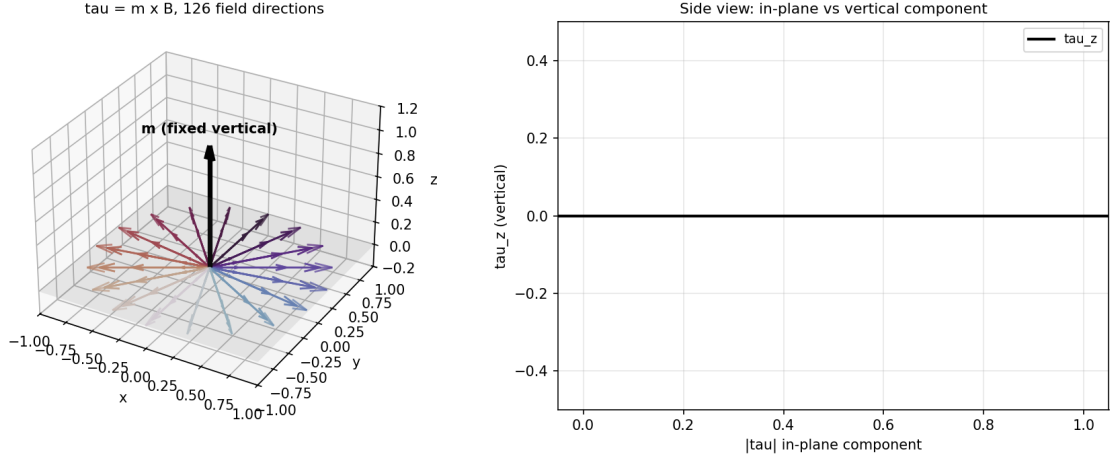


Figure 3.8: Torque under the point-dipole model is always perpendicular to the magnetic moment. The missing rotational direction is therefore a structural limitation rather than a controller-tuning failure.

trace and cannot form a minimum in all three translational directions. A restoring equilibrium in every direction is therefore unavailable from the static field alone.

Feedback is therefore required to provide the restoring force missing from the baseline schedule. Whether this is possible then depends on the directions in which the coil array can actually generate force and torque.

3.6 Actuator Authority

The available actuator directions were examined using the actuator-effectiveness Jacobian. It is shown that the leanr forces: F_x , F_y , and F_z can all be produced independently at the operating point.

The limitation appears in the rotational degrees of freedom when the pod is represented as a single magnetic dipole. For the point-dipole model,

$$\tau = \mathbf{m} \times \mathbf{B},$$

so the torque is always perpendicular to the magnetic moment. A torque about the magnet's own axis is therefore structurally impossible.

A singular value decomposition of the full $[F_x, F_y, F_z, T_x, T_y, T_z]$ Jacobian shows one zero singular value for every tested pose on the classic track preset. When the pod is level, the magnet axis matches the body z axis, so the unavailable direction is exactly straight up T_z . The remaining five directions are fully reachable.

When the pod is tilted, the unavailable direction rotates with the magnet and is distributed between multiple body axes. At $\theta = 0.2$ rad, a torque requested about a sideways axis is realised at approximately $\cos^2 \theta \approx 0.96$ on that axis, with $\sin \theta \cos \theta \approx 0.19$ appearing in T_z . Conversely, a torque requested about T_z produces only $\sin^2 \theta \approx 0.04$ on that axis, with approximately 0.19 coupling into the other rotational direction.

Thus, near normal level operation, yaw has almost no direct authority in the point-dipole model.

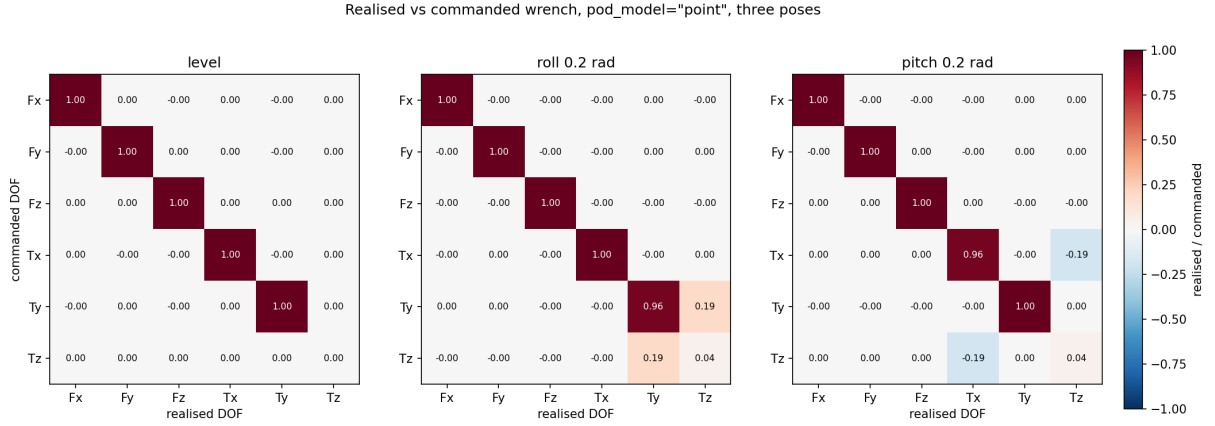


Figure 3.9: The realised-versus-commanded matrix under `pod_model="point"`. At level attitude, T_z is zero. At $\theta = 0.2$ rad, the unavailable direction is distributed between multiple axes.

This is a property of the model itself and cannot be recovered through controller gains.

3.6.1 Improvements in the Dipole Model

The original code implemented a single centroid dipole model. A new multi dipole model was implemented alongside this in the new modular code. In `pod_model="multi"`, the spatial separation of the magnets changes the available wrench directions. Because forces can now act at different positions on the pod, opposing forces can generate a moment about the vertical axis. The previous algebraic restriction therefore disappears and the full six-dimensional wrench becomes reachable.

All six directions reach 100% of their requested target. Five require approximately 0.5–2.5% of the current limit, while pitch (T_y) requires approximately 14.5%.

The point and multi-dipole results are therefore consistent with the base physics. In the point model, yaw is unavailable because a single dipole cannot experience torque about its own axis. In the finite-body model, forces acting at separated magnet positions can form a yaw moment.

Pitch, by contrast, was already reachable in the point model; its higher cost in the multi-dipole model instead comes from the geometry of the coil array. Producing a pitch moment requires opposing forces from coil rows ahead of and behind the pod, where the coils are further from the magnets and where changes in current also affect the levitation force. The actuator analysis therefore establishes that the new finite-body model provides full six-DOF wrench authority, but with a substantial difference in cost between directions. This is highly important for the weighting of future calibrated control systems.

3.7 Improvements in Control Architecture

The original simulator contained no feedback control. The modular code structure was therefore extended with three independent controllers sharing a common interface in `control/base.py`: lateral PD control for x/y force, vertical PD control for z force, and geometric SO(3) attitude PD control for roll and pitch torque. Each controller returns a `Wrench`. The combined controller sums the selected outputs, while the allocation layer converts the requested wrench into per-coil current corrections.

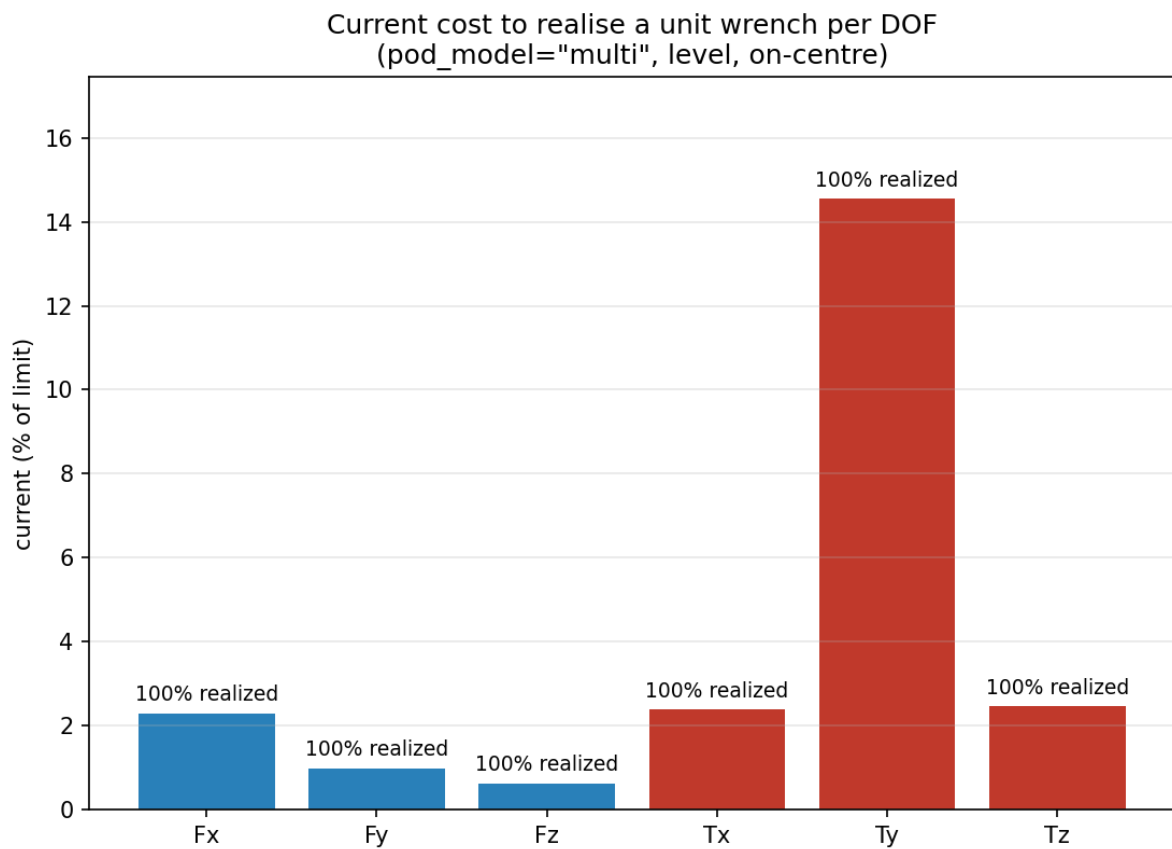


Figure 3.10: Full 6-of-6 rank recovered under the multi-dipole model. All six commanded wrench directions are realised, including yaw. Bar height shows the percentage of the available current limit required for each direction.

The stack was tested progressively from the same initial condition: baseline only, baseline plus lateral PD, baseline plus lateral and vertical PD, and the full controller stack. The resulting traces separate as controllers are added, with each law primarily affecting the axis it is designed to control. This demonstrates that the controllers are being applied and composed correctly.

The staged test used a 0.1 s window. During this test the original current was recalibrated every 5 ms so that its vertical force balanced the pod weight at its current pose. This was needed as the the current provided was far too little to balance/levitate the pod alone.

The short test duration is also important. The baseline’s lateral instability becomes apparent only after approximately 0.3–0.7 s. The 0.1 s staged test therefore ends before this instability develops. As a result of this though there is no hope of proving an established closed-loop stability.

The laws have two positive gain sets for the attitude controller. The gains used in the main simulation are illustrative rather than rig-tuned as these would need to be informed by the current in the coils.

The one longer closed-loop test isolates lateral authority over a 300 ms run from a 6.40 mm initial offset. It was run in three configurations that share every other setting (Figure 3.11):

1. *Vertical loop open* ($F_z = 0$, zero torque, uncalibrated baseline). The lateral error falls to 0.12 mm by 92 ms and reads 2.41 mm at 300 ms, but the pod is in free fall throughout, z dropping from 40.0 mm to -390 mm. The late-time value is not point control: it is the pod coasting laterally after it has fallen out of the useful field region, so most of the force it would need to correct against has gone. As §2.4.3 anticipated, this number cannot be separated from a free-fall artefact.
2. *Vertical loop engaged, shipped lateral gains* ($k_d = 5$; the baseline current re-levitated at the live pose every 5 ms, holding z at 40.0 mm to solver tolerance). The lateral error still falls to 0.19 mm by 64 ms, the actuators do produce the commanded corrective force, and the allocator realises it to within 1%. However, at $k_d = 5$ the loop is under-damped against the anti-restoring baseline stiffness. It overshoots and the error grows back to 5.16 mm by 300 ms, and diverges outright if the window is extended.
3. *Vertical loop engaged, $k_d = 20$* . With the height held and adequate damping the error decays monotonically to 0.8 mm at 220 ms, and then exhibits a light rise to 160 mm error at 300 ms. This means it is effectively countering the error in x from the original current allocation.

From 6.40 mm the loop reaches sub-millimetre error within 60–90 ms, so the coil array has real lateral control authority at the operating point and the allocator delivers it. The height-held runs also show that this is the *whole* of what the original experiment demonstrated. As with the free fall removed, the shipped gains do not hold still, and keeping the error small across the full window needs a higher k_d than the shipped set. This is consistent with the short-window caveat above: no closed-loop stability is established, only local corrective authority.

The shipped [ClosedLoopFull] gains, selected through `main.py` option 7, were not rig-calibrated. They were sized using the simulator’s measured actuator force and torque limits:

Adding the vertical and attitude controllers does not overcome the main limitation because the underlying baseline schedule remains anti-restoring in x and weak in y and does not provide sustained levitation. Once the pod moves away from the central trim region, the positive lateral stiffness drives it further from the centre while the loss of height removes it from the useful field region. However, K_z was assigned by inspection from Figure.3.5, the preset x offset was

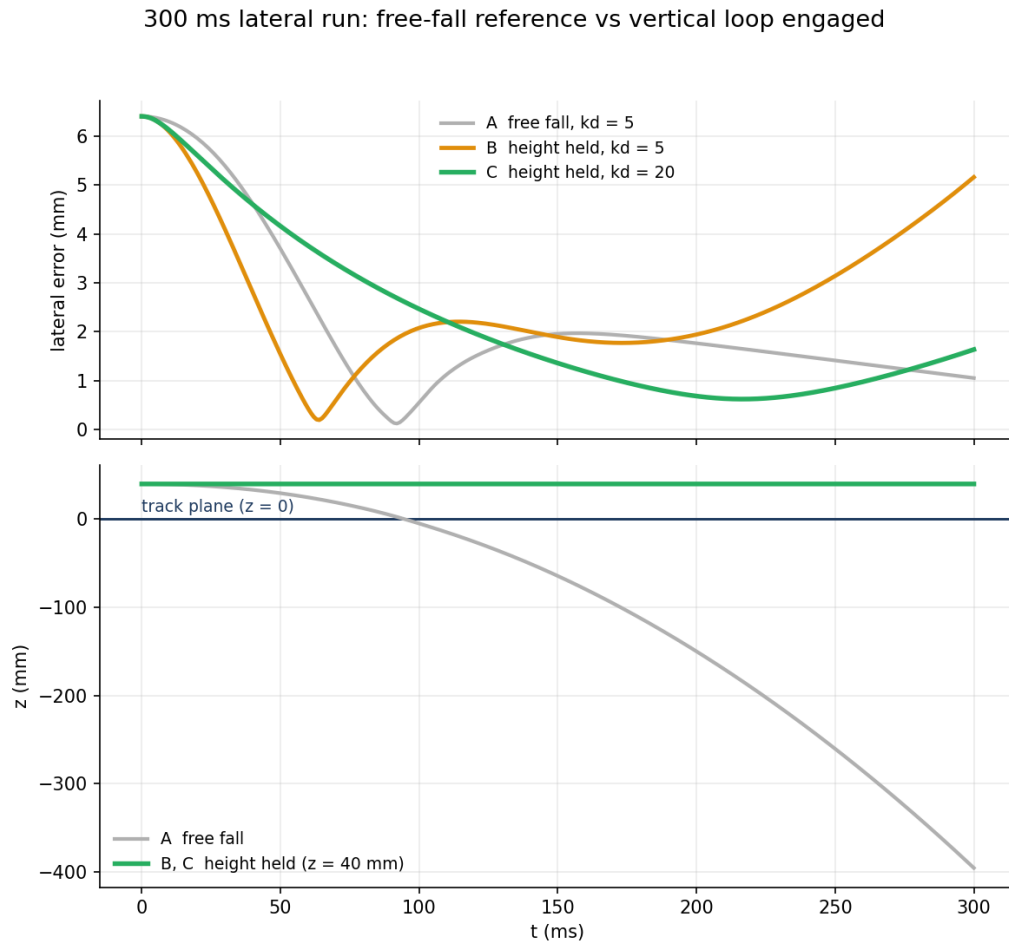


Figure 3.11: The 300 ms lateral run in three configurations from the same 6.40 mm offset. With three variations of experiment run, a baseline with no z restoring term in freefall, and two vertically held pods with varied lateral stiffness constants

6.40mm, which was read from the lateral stiffness. The 300 ms lateral run in three configurations from the same 6.40 mm offset. The shared motion feedback shows the effect of the tighter tuning with the higher k_d value. The simulation breaks with a time span greater than 400ms as the lateral error increasing to around 5000 fold by 1s.

The control results demonstrate that the modular controllers can be applied, composed, and allocated to the available actuator directions, and that the lateral controller can reduce lateral error. They do not demonstrate stable levitation because the original does not allow for the operating conditions required to test this.

3.8 Results summary

- The modularisation preserves the original simulator’s behaviour, demonstrated by the end-to-end test and the zero numerical difference in the selected to test copied kernels.
- The new $SO(3)$ attitude representation removes the large-rotation failure of the Euler-angle representation while remaining in close agreement for the small rotations relevant to normal operation.
- The investigation identifies the inherited baseline schedule as the main barrier to stable levitation. Its vertical force is insufficient without calibration, its lateral stiffness is anti-restoring, and its operating region is therefore not a stable equilibrium. Feedback is consequently required, but the controller cannot overcome an unsuitable baseline as it has the wrong sign.
- The new actuator and control components demonstrate new capabilities. The point-dipole model has a structural five-DOF limitation, while the multi-dipole model restores full six-DOF authority. The controllers can be composed and allocated through the new architecture, and lateral feedback produces a real corrective response at the operating point, the error is driven below 1 mm within ~ 70 ms with the height held. Stable levitation and controlled forward travel, however, remain un demonstrated because the inherited baseline does not provide the sustained, stable operating point required for those experiments.

The main contribution is therefore the validated modular architecture and the capabilities it makes possible, rather than a completed levitation controller. The new physical and control components establish a basis for future work in which the inherited current schedule can be replaced or re-fitted and the resulting closed-loop system can then be tuned and experimentally validated.

Chapter 4

Discussion

The aim of the project was to refactor the code and to provide test cases and infrastructure in order to facilitate enhancements to the model. This was not fully accomplished and tested until late into the development cycle.

4.1 Reporting against the objectives

Chapter 1 set four objectives. Three were met in full and one was met only in part:

Objective 1: faithful reconstruction was met. The monolithic script was decomposed into seven dependency-ordered packages, and the reconstruction was able to reproduce the original codes behaviour at runtime rather, additionally the six copied kernels agree to a maximum absolute difference of exactly zero (Table 3.1), and all twelve state components agree across an end-to-end trajectory to 0.0 throughout a developing instability (Figure 3.1).

Objective 2: establishing the reachable wrench directions was met for reachability, partly for strength. The three translational directions are independently reachable at the operating point. Under the inherited point-dipole pod model the rotational block is rank-deficient by owing to yaw. The multi-dipole model recovers the sixth direction through the same coils and with no hardware change (Figure 3.10). The second half of the objective: whether those directions are *sufficient* for stabilisation is answered in the negative, and for a reason unrelated to rank: reachability fixes only which directions a force can be produced in, not whether the environment those forces act against can produce a stable operating point. It does not, for the reasons given in §3.4.

Objective 3: a closed-loop layer separating wrench generation from current allocation was done architecturally and demonstrated in part. The three control laws compose through a common wrench representation and the allocator realises the commanded force to within one per cent of target. The experiments establish corrective authority, rather than keeping to one point. This is unasailable without a new baseline control, not necessarily the architecture §4.7 separates the two.

Objective 4: the range of the Euler-angle formulation was met. The two representations agree to better than a milliradian out to roughly 10° of total rotation excursion and diverge rapidly beyond it (Figure 3.3), with the SO(3) conserving angular momentum to a relative error of $\sim 10^{-13}$ where the Euler-angle path does not. The practical consequence is drawn in §3.3.

4.2 Outcome of the Refactor

The benefits of the refactor became clear once the code was used to investigate the research questions. After parity with the original implementation had been established, individual parts of the simulation could be changed without having to work through the whole codebase. For example, the attitude integrator study required changes only to the integration functions, while the control-law study added new controllers between the existing current calculation and coil-current generation. In both cases, the field model and baseline current calculation remained unchanged.

The work on the control process also benefitted from the refactor.. The `Controller/Wrench` model improvements allowed for the implementation of the three described control strategies without changing the underlying simulation, while `physics/field_query.py` kept the inherited field calculation separate from the control architecture. These boundaries were useful because they corresponded to parts of the system that genuinely needed to be changed independently.

This is strong evidence for the importance of decomposition and modularisation: parts of a system are separated so that changes do not infect other areas⁴⁵. The parity test was important here because it showed that the new modular code still behaved identically to the original. This gave a reliable starting point for the later investigations.

Not every boundary imposed was equally useful, however. Some of the decomposition was carried further than what was strictly needed. The original package structure was developed with the general aim of making the inherited code manageable and testable, before the later research questions were fully known. The structure was therefore not designed specifically around the investigations that eventually appeared in Chapter ??, or indeed what future work would need to be done. This is consistent with the observation that real software designs are rarely produced from a completely known set of requirements⁴⁶.

The result was a substantially more structured codebase, with 1,861 lines of original code re-organised into the seven-package architecture described in §2.3.2. This made the simulation much easier to extend, but some of the resulting structure added more complexity than benefit. The `configuration/` package, for example, eventually had to be simplified to a single module because the separation made it unclear which file needed to be changed for a particular simulation.

This highlights that the success of a decomposition should not be measured by the number of modules it produces, as in many cases this will just confound future developers and not aide in the extensibility and understanding that motivated the work in the first place. A useful boundary is one where there is a clear reason that the component may need to change, be tested or be replaced independently⁴⁵. The later investigations provided exactly this information: they showed which boundaries were genuinely useful and which had introduced unnecessary complexity.

This also suggests a practical direction for future development. The architecture should now be treated as a working foundation rather than as something that needs to be decomposed further. Future changes should favour simple work around components that have demonstrated a need to be improved by fully motivated physics, rather than creating additional modules simply to make the structure appear more modular.

4.3 Validation of Refactor

The parity test should not be interpreted as a validation of the physics. It shows that the restructuring preserved the behaviour of the original model. The original model still contains its existing physical and numerical limitations, which were deliberately left unchanged during the reconstruction. The first objective was to reproduce the inherited system rather than correct it.

The perfect agreement through the unstable test case in Figure 3.1 is particularly useful in this respect. Both implementations use the same numerical kernels, integrator, so the test checks whether the new modular route through the code produces the same result as the original route. It is therefore strong evidence that the refactor did not introduce new discrepancies, but it does not establish that the underlying numerical or physical model is correct.

This becomes important when interpreting the later results. Once the reconstruction had been validated, weaknesses in the inherited model could be investigated separately rather than being confused with errors introduced by refactoring.

4.4 Pod Geometry Improvements

The excursion into quaternion/SO(3) dynamics in an attempt to replace Euler angles was overlooking the more direct problems at hand. Whilst the modularisation made this easy to amend, this did not necessarily need to change.

Posability of the pod and its tilt are highly important - as demonstrated by the bleed exhibited at values of roll and pitch - but this did not make them the largest points of error. All disturbances can be taken as small angle approximations as any tumbles or singularities occurring would have had catastrophic impacts on any real pod.

The SO(3) integrator avoids the singularity associated with Euler angles and provides substantially better long-term preservation of the relevant geometric structure. The remaining attitude work is therefore less about establishing whether the method works and more about deciding which integration method should become the standard production path.

4.5 Actuator Authority Variability

The actuator-authority results show a limitation with the single point dipole model. The limitation is not just that the coil array cannot generate the forces of the required magnitude. A greater issue is that some torque forces are not reachable. The three translational force directions were all independently mapped: however, this was not the case for rotational directions when the pod is represented as a single point dipole.

The multi-dipole model changes the physical representation rather than simply changing the controller. By representing the individual magnets at separated positions, forces can act away from the pod centre and produce moments through their different lines of action. The multi-dipole result demonstrates that the spatial structure of the pod provides the missing authority.

The accuracy of this multi-dipole model change is not independently validated in this report, and needs further experimentation. This validation is an important question to ask of the work: if this is not feasible then controlling yaw with the current track layout is impossible.

However, full reachability does not mean that all control directions are equally easy to produce. Pitch requires substantially more current than the other tested wrench directions. This disparity

in the allocator map matters greatly for control law construction, as was discussed with respect to the z component³⁹. An allocator that treats all six directions equally could use the available current inefficiently or approach actuator limits in directions that are expensive to produce.

There is also an important qualification to authority as demonstrated in the example. The initial experiment was carried on the classic track. The pattern greatly effects the fields produced and the granularity. The present authority result therefore establishes reachability for the tested actuation model. All of the tracks are mirror symmetric about the centre line so the torque they can exert is an open question at this time.

4.6 Track Structure Improvements

Breaking away from the 2D planar model could be the answer for the track geometry redesign. This is a substantial future problem, not attempted by this analysis. The major roadblock to this is access to experimental data. The coil currents need to be physically demonstrated which requires the model to be made and the physical tests to be run.

The present implementation uses the experiments which have happened on the fixed 2D geometries from the .xlsx files, which correspond to the Classic and Hex layouts. This could likely be replaced by a geometry generator. Having a program which generates spacing, magnet position, and magnet orientation would allow future work to build the geometry rather than hand-specify it. Additionally this would enable 3D configurations, greatly enabling access to the rotational degrees of freedom.

This does not, however, change the fundamental control requirement. The saddle-point instability identified above follows from the pod's fixed magnetic moment and the current-free equilibrium, regardless of the particular coil arrangement. Track geometry can improve the cost and conditioning of producing a given wrench, but it cannot remove the need for active feedback.

Taken together, these results demonstrate that full six-DOF authority is *possible* under the multi-dipole, per-magnet-actuation idealisation; they do not demonstrate that this authority is *achievable* on the track as specified. Establishing that distinction requires the actuator grouping, pod model, and operating envelope to be validated together.

4.7 Control Laws

The modular control architecture successfully separates the control mechanism from the original current schedule. The new lateral, vertical and attitude controllers use a common wrench-based tooling, while the allocation layer converts the requested wrench into the desired coil-current corrections. This allows different control laws to be tested without rewriting the original baseline.

The staged tests show that these components behave as intended. When individual control laws are introduced, the corresponding state variables respond in the expected directions, and the controllers can be combined through the common wrench representation. This provides evidence that the control architecture is correctly integrated with the reconstructed simulator.

The longer lateral-control experiment gives the clearest demonstration of *transient* control authority. From a 6.40 mm offset the lateral loop drives the error below 1 mm within 60–90 ms, and it does so whether or not the height is held, with the allocator realising the commanded force to within one percent. This shows the available actuators can generate a corrective lateral response at the operating point. It does not show sustained lateral control: once the confounding free fall of the original run is removed, the shipped gains overshoot and the error grows back

over the 300 ms window, and holding the error small to 300 ms needs a higher derivative gain than the shipped set. The experiment therefore establishes local lateral authority, not point control.

However, the results of the control laws go little further than small architectural improvements. The pod fell in all experiments as the standard current schedule could not support it vertically. More importantly, the current ruling remained laterally unstable. The experiment therefore demonstrates local lateral control authority, rather than sustained levitation. The work for the attitude control investigated the impacts of these newly developed PID inspired controllers had no real impact on the results. No stability could be ensured because the results are cumulative rather than replacing of the baseline schedule.

To calibrate the control gains properly needs careful alignment with experimental coil data. This integration was not carried out in this development phase, owing to the primary focus being the architectural reformation of the simulation. A future student with physics and experimental background can perform this individual model tuning, made possible by the construction of the base classes.

This distinction is central to interpreting the closed-loop results. The controller can oppose the lateral error, but it cannot create a viable levitation operating point when the original simulation does not provide one. As the pod loses height, it also leaves the useful field region, making the lateral controller increasingly difficult to evaluate.

The failure of the longer run is therefore not evidence that the lateral controller itself is ineffective. Instead, it demonstrates the practical limit of the original current allocation method. This is why controller tuning should happen after the replacement of the baseline schedule.

Beyond PID controllers, other control laws were discussed. Implementing these more complex rules would likely perform well as PID controllers exist multiple layers of abstraction below this³⁴.

4.8 Remaining Model development

The control results expose several areas where the inherited physical model needs to be developed before the simulation can support stronger real world claims.

The first is induced-current damping. The current model contains no eddy-current damping, meaning that the simulated pod does not have a mechanism corresponding to this physical source of damping. This becomes particularly important for any future investigation of electrodynamic behaviour or longer-duration motion. A suitable damping model should therefore be introduced and validated before such results are treated as representative of a physical system.

The inertia model should also be improved. The current inertia tensor is simplified rather than being derived from measured geometry or a detailed CAD model. This is unlikely to dominate the translational results as the code is still far away from fully explaining or informing experimental behaviour, but it becomes increasingly important once attitude control is being investigated. A measured or CAD-derived inertia tensor would therefore be a straightforward improvement before attempting detailed attitude-controller tuning.

The field model also requires experimental validation. The magnetometer measurements were not applied in this development cycle as the work was fundamentally architectural rather than being truly physics informed. We cannot fully establish model accuracy because the drive current associated with the measurement was not recorded. A future experiment should therefore record the coil current together with the field measurement. This would allow the simulated and

measured fields to be compared under known excitation and would provide a much stronger basis for judging the physical accuracy of the inherited field model.

These improvements should be treated as separate from the software-parity result. The purpose of the reconstruction was to preserve the inherited model and allow extension, while the purpose of the next stage is to determine which parts of that model need to be replaced with physically validated alternatives.

4.9 Limitations of Applied Method

It has become apparent throughout the course of this work that several fundamental shifts needed to happen to the NIFTI simulation code to allow it to be used for further development. The original monolithic code was too complex to comprehend inline and modify. Under the surface there were unexpected behaviours of some features, and programming choices which were only questioned after it had been reproduced. A more iterative approach to the development would have resulted in several of these features being addressed earlier.

The decision to perform the work without using the specific experimental coil values was an oversight in the fact that none of the tuning work was performed. Whilst this is not fundamentally limiting for the architecture of the system, it leaves a complex process for the future students as these new systems have to be carefully tuned.

Fundamentally not agreeing the work with the outside data was the limitation of the project, whilst the new control laws were implemented any form of true validation could not occur for them as they were fundamentally limited by the baseline schedule

Much of the lack of extensive result data is caused by two reasons. Firstly the modularisation and accompanying test suite did not yield any results for direct analysis. Secondly many of the extensions implemented were limited owing to the decision to make the project more architectural and diagnostic than based in the pure physics of the system. This has, however, clarified the modelling and simulation work that is needed to make this project move forward.

Chapter 5

Conclusion

The primary output of this work is the reconstructed simulation. A single 1,861-line script, in which the field model, geometry, integrator and current assignment were interwoven, was rebuilt as 33 modules in seven dependency-ordered packages (§??). Each package was extracted and run using only packages that had already been extracted and verified. A provenance record was kept as each function was moved, while per-package regression tests were carried out during extraction. This meant that discrepancies could be localised to a single move or error rather than appearing as an unexplained difference in a full simulation run. Given the complexity of the simulation, even a small implementation difference could produce a markedly different trajectory, so exact agreement was required at every step rather than agreement within a tolerance.

Two kinds of evidence establish that the reconstruction succeeded. Six copied kernels agree with the original to a maximum absolute difference of zero, and all twelve state components (Cartesian coordinates, Euler angles and associated forces) of an end-to-end trajectory agree throughout a developing instability (Figure ??). Each of the three extensions entered the codebase through a single boundary: the attitude integrator changed only the integration functions, the control layer was inserted between the existing current calculation and coil drive, and the finite-body model replaced the pod force term, with the field model and inherited schedule left untouched in each case. A decomposition is only really tested when something new has to be fitted into it, and these boundaries held.

Not all of the extensions were necessary. The structure was planned to make the inherited code manageable before the research questions were known. The useful measure of a decomposition, however, is not how many modules it produces but whether each boundary marks something that genuinely needs to change, be tested or be replaced on its own. From this, the architecture should be treated as a working foundation rather than something to subdivide further.

That foundation made possible a set of findings that could not previously be attributed with the same confidence. In the inherited script, a defect in the physical model and a defect in the code evaluating it were indistinguishable; with the consistency of the reconstructed codebase confirmed, they are now clear. The inherited current schedule cannot levitate the pod: its raw peak lift is approximately 3.6 mN against a pod weight of 1.71 N, and its lateral stiffness at the operating point is positive, so it drives the pod away from the centre (§??). The first is a question of scale and was worked around; the second is a question of sign and cannot be recovered by any control law as written. Within that boundary, the new components behave as designed: lateral feedback reduces a 6.40 mm offset to below a millimetre in 60–90 ms, while the actuator analysis shows that the point-dipole pod model is structurally limited to five of six wrench directions. The multi-dipole model recovers the sixth through the same coils, although this result remains unvalidated against measurement.

Re-fitting the schedule against a restoring vertical equilibrium is therefore the precondition for addressing the project's remaining questions. Once that is done, however, the reconstructed architecture provides a way to test each new component independently without disturbing the rest of the validated system.

Chapter 6

Future Work

The remaining work has one key roadblock before other avenues open up, that is having a levitation baseline that defines a stable operating point in height of the pod.

6.1 A credible Levitation Scheme

The inherited current schedule is the blocking limitation identified throughout Chapters 4 and 5: its raw lift is hundreds of times below the pod weight and its x stiffness pushes the pod away. The first task is therefore diagnostic: sweeping coil currents to locate a restoring vertical equilibrium. This would be followed by re-fitting the schedule with a lift law that holds that equilibrium without per-step recalibration. Until this exists, closed-loop runs have no stable point to hold and station-keeping cannot be tested.

6.2 Rig-data controller tuning

The PD gains reported here are illustrative, sized from the simulator's own force and torque limits rather than from hardware. Once a stable baseline exists, each coil group should be driven with a swept current about the operating point and the pod response recorded; from this the pod's effective mass and inertia and the schedule's local stiffness can be fitted, the PD gains (currently illustrative) set for a chosen closed-loop bandwidth and damping, and the result refined on the rig against step and disturbance tests. The achievable bandwidth is capped by the coil amplifiers and the position sensors.

6.3 Finite-body model in the integrated simulation

The multi-dipole model exists only within notebooks, not the wider simulation or library. Placing it into the main closed-loop path would allow the controller to exploit the six-degree-of-freedom authority, that the point model cannot reach. The multi-dipole change itself also needs independent validation: if the separated-magnet torque cannot be demonstrated experimentally, yaw control on the present track layout is not achievable.

6.4 Actuator-authority pose sweeps

The reported authority and cost figures are single-pose snapshots at one level and one tilted attitude. The modular sweep capability should be used to map authority and per-direction cost across the full range of tilt poses and across track presets. This is computationally expensive, requiring the construction of a full field and Jacobian evaluation per pose but is now mechanically straightforward.

6.5 Track-geometry generation and 3D layouts

The present geometries are hand-specified from the inherited `.xlsx` files. A generator producing coil spacing, magnet position and magnet orientation would let future work build geometries rather than transcribe them, and would open the door to 3D configurations that give direct access to the rotational degrees of freedom. Any new geometry still requires experimental coil-current data before its results can be trusted, and no geometry removes the underlying active-control requirement.

6.6 Physical-model development

Two inherited simplifications should be revisited once they materially affect the control problem. The model currently contains no induced-current (eddy-current) damping, which as shown in the literature has a marked effect of maglev vehicles and would likely play an important part in longer-duration simulations because it can provide passive damping that is absent from the present model. The inertia tensor is also simplified rather than measured or CAD-derived; a measured tensor would be a straightforward improvement before detailed attitude-controller tuning.

6.7 More Advanced Control Loops

The slottable controller interface was designed to accept alternative control laws. The most promising direction is a reinforcement-learning stabiliser, though realising it will likely require overturning the baseline schedule and reworking the surrounding architecture rather than adding a law on top of it.

Underlying all of these is the single largest gap in the present work: the simulation has not been brought into agreement with the external physics of the real system. The reconstruction was deliberately architectural in scope rather than physics-informed, and closing that gap is the precondition for any real-world claim about NIFTI.

Appendix A

Allocator Jacobian: Exactness Verification

The allocator inverts an exact per-ampere map from actuator-group currents to the wrench they produce⁴⁷. Its j -th column is the wrench from one ampere through actuator group g_j ,

$$J \in \mathbb{R}^{6 \times n_g}, \quad J_{:,j} = \left(\sum_{c \in g_j} \mathbf{m} \cdot \nabla \mathbf{B}_c, \quad \mathbf{m} \times \mathbf{B}_{g_j} \right), \quad (\text{A.1})$$

built once per pod pose and reused. A target wrench $\mathbf{w}^*$ is mapped to a current correction by a weighted minimum-norm solve $\delta \mathbf{I} = (WJ)^+ W \mathbf{w}^*$, with the diagonal weight W set to preserve the vertical force and to suppress torque about the pod's own moment axis, and each group current then clipped to $[0, I_{\max}]$.

This map is exact rather than a local linearisation, because a coil's field is exactly proportional to its drive current. That exactness is verified three ways, all in `tests/test_control.py`:

1. **Two-way wrench construction.** For a set of non-unit test currents across the real track, the resulting wrench is computed once as $J \mathbf{I}$ and once by directly evaluating the field and gradient sums through the field model, and the two are required to agree.
2. **Superposition.** The wrench from currents $\mathbf{I}^{(1)} + \mathbf{I}^{(2)}$ equals the sum of the wrenches from $\mathbf{I}^{(1)}$ and $\mathbf{I}^{(2)}$ separately.
3. **Homogeneity.** Scaling all currents by α scales the wrench by exactly α .

Force and torque agreement are reported separately. The torque block of (A.1) is a closed-form cross product and agrees to near machine precision; the force block depends on a numerically differenced $\nabla \mathbf{B}$ (central differences, $h = 10^{-5}$ m) and carries a correspondingly larger but still small finite-difference residual. A locally valid linearisation would pass none of the three checks away from its expansion point; an exact linear map passes all three at any current.

Appendix B

Field-Kernel Implementation Details

This appendix contains implementation-level detail relevant to reproducibility but disproportionate to the main narrative (§2.1.1, Background; it is referenced from there and not needed for the main argument.

B.1 Closed-Form Field Solution

Every magnetic source is modelled as a finite right cylinder of radius a and half-height b , carrying a uniform effective magnetisation M (A/m). For a permanent magnet, M is the material's intrinsic magnetisation; for an electromagnet, $M = NI\mu_r/L$ with N turns, current I , relative permeability μ_r , and length L .

The finite cylinder is treated as the sum of two boundary contributions, one at each end face ($z = +b$ and $z = -b$). For a field point at cylindrical radius ρ and axial offset z :

$$B_0 = \frac{\mu_0 M}{\pi}, \quad z_+ = z + b, \quad z_- = z - b$$
$$\alpha_{\pm} = \frac{a}{\sqrt{z_{\pm}^2 + (\rho + a)^2}}, \quad \beta_{\pm} = \frac{z_{\pm}}{\sqrt{z_{\pm}^2 + (\rho + a)^2}}, \quad \gamma = \frac{a - \rho}{a + \rho}, \quad k_{\pm} = \sqrt{\frac{z_{\pm}^2 + (a - \rho)^2}{z_{\pm}^2 + (a + \rho)^2}}$$

$k_{\pm} \rightarrow 1$ as the field point approaches the coil's own edge ($\rho \rightarrow a$, $z \rightarrow \pm b$) which is why the elliptic-integral evaluator clamps its modulus argument near this boundary rather than diverging.

B.2 The Generalised Elliptic Integral

Rather than implementing the three classical kinds of complete elliptic integral separately, the code uses Bulirsch's `cel(kc, p, c, s)` algorithm:

$$\text{cel}(k_c, p, c, s) = \int_0^{\pi/2} \frac{c \cos^2 \phi + s \sin^2 \phi}{(\cos^2 \phi + p \sin^2 \phi) \sqrt{\cos^2 \phi + k_c^2 \sin^2 \phi}} d\phi$$

evaluated by an arithmetic–geometric–mean–style duplication iteration, converging rapidly rather than by series expansion. Different (p, c, s) triples yield the different combinations each field component requires — e.g. `elliptic(k, 1.0, 1.0, -1.0)` for the radial component versus

`elliptic(k, gamma2, 1.0, gamma)` for the axial component — without maintaining three separate special-function implementations.

The radial and axial field components are the weighted difference of the routine at the top- and bottom-boundary moduli:

$$B_\rho = B_0 (\alpha_+ \text{cel}(k_+, 1, 1, -1) - \alpha_- \text{cel}(k_-, 1, 1, -1))$$

$$B_z = \frac{B_0 a}{a + \rho} (\beta_+ \text{cel}(k_+, \gamma^2, 1, \gamma) - \beta_- \text{cel}(k_-, \gamma^2, 1, \gamma))$$

converted to Cartesian components via the field point’s azimuth, $B_x = B_\rho \cos \phi$, $B_y = B_\rho \sin \phi$. The field of an entire collection (track or pod) is the plain vector superposition of this single-cylinder result over every magnet it contains — linear in the number of sources, with no cross-terms, which is the property §2.1.1 and Background §2.4.1 depend on.

B.3 Numba JIT Compilation

Exactly four functions in `physics/field.py`: `constants()`, `elliptic()`, `B_rho()`, `B_z()` are decorated `@numba.njit`: the innermost per-(point, magnet) evaluation of the closed-form field solution. Numba’s JIT requires these to remain plain, module-level functions with no dependency on Python objects it cannot compile, which is why they sit at the bottom of the physics layer. These four functions are unmodified from the original `baseline_code.py` implementation

The outer grid/point/magnet loops in `simulation/potential.py` remain nested plain-Python `for` loops, not vectorised or JIT-compiled, again inherited from the original code, and the natural next target if sweep runtime becomes limiting in future work.

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